

Optimal Transport for Complex Sharing Economies with QoS Guarantees

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Abstract

This dissertation addresses the complex dynamics of resource allocation in contemporary sharing economies (SEs), which feature stochastic, competing demand groups and hybrid supply sources. Conventional allocation methods fail to leverage the full probabilistic distributions, leading to inefficient and suboptimal outcomes, increasing costs for operators. This research proposes and develops a novel framework based on **Fully Probabilistic Design (FPD)** and **entropy-regularized Optimal Transport (OT)** to generate cost-optimal, interpretable, and Quality of Service (QoS) guaranteed allocation plans. This approach is particularly relevant for SE scheme designers seeking more sophisticated decision-making tools. The FPD-OT framework is systematically applied to SE scenarios of increasing complexity, culminating in a model that accommodates **bivariate competing demands** and **bivariate hybrid supplies**. A key contribution is the explicit integration of QoS guarantee, constraining the probability of undersupply, into the FPD-OT optimization. This is achieved through a feasibility analysis to determine minimum achievable undersupply and a **modified Sinkhorn-Knopp** algorithm designed to enforce these QoS constraints while respecting the supply and demand distributions. Comparative evaluations against state-of-the-art allocation strategies are conducted using Monte Carlo simulations. Results demonstrate that FPD-OT derived strategies achieve superior performance in minimizing operational costs. The research provides SE scheme designers with a robust methodology to develop adaptive and probabilistically coherent allocation policies.

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1 Notation

Table 1: Key Mathematical Notations Used in the Dissertation

Symbol	Meaning
x_1, x_2	Quantities from two distinct supply sources (hybrid supplies)
y_1, y_2	Quantities for two distinct demand groups (competing demands)
$\mathbf{x}$	Supply vector. Discrete resource quantities from distinct supply sources. Univariate supply, $\mathbf{x} = (x_1)$ and for bivariate supply, $\mathbf{x} = (x_1, x_2)$
$\mathbf{y}$	Demand vector. Discrete resource quantities from distinct demand groups. Univariate demand, $\mathbf{y} = (y_1)$ and for bivariate demand, $\mathbf{y} = (y_1, y_2)$
$Q_{\max}$	Maximum possible quantity for total supply or demand
Ω_S, Ω_D	Discrete sets of possible supply and demand quantities
$\mu(\mathbf{x})$	Supply marginal probability mass function over Ω_S
$\nu(\mathbf{y})$	Demand marginal probability mass function over Ω_D
$C(\mathbf{x}, \mathbf{y})$	Cost function for matching supply $\mathbf{x}$ with demand $\mathbf{y}$
c_o	Unit cost parameter for oversupply
c_u	Unit cost parameter for undersupply
c_t	Unit cost parameter for supply exceeding threshold
x_{thresh}	Supply threshold for the c_t cost component
ϵ	Entropy regularization parameter (controls smoothness/cost influence)
$\phi(\mathbf{x}, \mathbf{y})$	Base Distribution (often called reference measure)
$\pi^I(\mathbf{x}, \mathbf{y})$	Ideal distribution
$\pi(\mathbf{x}, \mathbf{y})$	Joint probability distribution / plan
$\pi(x_1, x_2, y_1, y_2)$	4D Joint probability distribution (bivariate supply with bivariate demand)
$KL(\pi \pi^I)$	Kullback-Leibler divergence from plan π to ideal plan π^I
$\pi^O(\mathbf{x}, \mathbf{y})$	Optimal joint probability distribution
Δ^{nm-1}	Probability simplex representing all possible $n \times m$ joint probability plans
$\Pi(\mu, \nu)$	Set of all feasible plans, $\pi^O(\mathbf{x}, \mathbf{y})$ satisfying marginals $\mu(\mathbf{x})$ and $\nu(\mathbf{y})$
α	Quality of Service (QoS) parameter, often $P(\text{Undersupply}) < \alpha$
$P_\pi(x_1 + x_2 < y_1 + y_2)$	Probability of total undersupply in a plan $\pi(\mathbf{x}, \mathbf{y})$ with bivariate demand and bivariate supply
$\pi^O(\mathbf{x} \mathbf{y})$	Optimal conditional probability distribution of supply $\mathbf{x}$ given demand state $\mathbf{y}$
$\pi^O(\mathbf{y} \mathbf{x})$	Optimal conditional probability distribution of demand $\mathbf{y}$ given supply state $\mathbf{x}$
$E_\pi[C(\mathbf{x}, \mathbf{y})]$	Expected cost under a given joint plan π
$\pi_{QoS}^O(\mathbf{x}, \mathbf{y})$	Optimal joint plan satisfying specified QoS constraints
$\alpha_{\min}$	Minimum achievable probability of undersupply
λ_{kMc}	Mean parameter of Poisson distribution: $k \in \{s, d\}$: Supply or Demand. M (e.g., 1, 2): Identifies the specific demand/supply group. c (e.g., a, b): Identifies the mode number of the distribution.
p_k	Mixture weight of the two modes of a Poisson Distribution. $p_k \in [0, 1]$

2 Introduction to Sharing Economies

Since the 2010s, the collaborative economy has emerged as a pivotal challenger to conventional resource markets. Sharing economies (SE) are peer-to-peer networks which leverage underutilized human or physical resources [1]. SEs depart from traditional models predicated on individual ownership and centralized control, separating **resource ownership** from **resource utility**. For example, ride-sharing apps use idle cars and drivers (resources), for customers transport (utility). Buying mobility, not a vehicle, exemplifies SEs shifting focus from ownership to access. Examples include Uber (ride-hailing), Airbnb (homestays), and Roadie (gig workers logistics).

A key enabler and, simultaneously, a core challenge is the inherent flexibility within these systems. Providers operate autonomously, contributing resources based on convenience and incentives, and consumers seek services on demand [2]. This decentralisation and dynamism makes supply and demand inherently **stochastic processes**. Availability and intensity fluctuate with combination of geography, weather, events, and human behaviours, introducing uncertainty into platform operations [3].

2.1 Rise of Multivariate SE Supply and Demand

The facilitators of SE schemes (**scheme designers**) follow asset-light models to match supply and demand. Historically, SE schemes dealt with matching singular supply (e.g., drivers, EV chargers) with singular demands (e.g., customers, cars). Modern platforms are more complex [4]:

- **Competing Demands:** Diversified user groups create multiple demand segments for the same resource within a platform. For example, transport SE schemes serve tourists and daily commuters; others segment demand segmented by location or price sensitivity. Ignoring these multi-dimensional correlated demands causes inefficient allocation, missed revenue, and risks of systematically underserving certain user groups [5].
- **Hybrid Supplies:** Similarly, supply sources are also diverse. Inventory can be a combination of platform-owned and partnerships [6]. Car rentals may offer Electric and Combustion vehicles to the same users [7]. “Surge Supplies” (primary and secondary, with secondary access only during demand spikes [8]) have been proposed to counter dynamic pricing.

These multidimensional supplies and demands might have inherent correlations too (e.g., rain increases ride demand for tourists and commuters). Resource shortages (shocks) can affect all supply sources simultaneously. This evolution towards multivariate, correlated stochastic demands and supplies, renders traditional resource allocation strategies increasingly inadequate. Decision-making in such complex, dynamic, and uncertain systems is challenging.

2.2 Decision Making in SEs: State of the Art

Extensive research addresses SE resource management using diverse methodologies. This classifies approaches by their handling of uncertainty (deterministic to probabilistic), with supply and demand as probability distributions.

2.2.1 Deterministic and Heuristic Approaches

The following methods abstract away the stochastic nature of SEs. Their inputs and outputs are single-value “contracts” used irrespective of the uncertainty of supply and demand.

1. **Fixed Quotas and Rules:** Static allocation rules such as simply assigning a number of vehicles per zone or a First Come First Serve (FCFS) method lack adaptability for dynamic environments and raises fairness concerns [8].
2. **Classical Optimisation (Linear Programs):** Conventional Mixed-Integer Linear Programs typically take in fixed deterministic inputs such as historical averages, whose observation probability may be low.

These methods lack robustness and efficiency, increasing costs or missed sales for the scheme designer, especially in dynamic markets.

2.2.2 Probabilistic Single-Value Decisions

Several SE schemes acknowledge uncertainty by using probabilistic inputs but yield a single fixed decision, optimal in an “expected” sense.

1. **Mean Matching:** Allocates resources based on mean or quantile (e.g. 95th percentile) of the historical demand distribution. These fixed, probabilistically informed supply decisions may fail during common surges (e.g., NYC taxi study showed even 17%-38% surges degraded service drastically [9]).
2. **Newsvendor Model:** This archetypal single-value probabilistic model uses the full demand distribution and certain cost parameters to prescribe one optimal quantity to supply. Primarily capturing economic tradeoffs, the standard form is insufficient for multivariate, correlated SE problems. [10]
3. **Sample Average Approximation (SAA):** This method samples from probability distributions and solves an optimisation problem to minimize a certain cost returning a single point value [11].

These methods represent the uncertainty within demands and supplies but single contracts discard valuable distributional information. Scheme designer needs more than a single contract. They require a **plan** to understand the potential risks and opportunities over a distribution of optimal contracts.

2.2.3 Dynamic and Control Based Systems

Extended set of methodologies focus on real-time adjustments and strategies but can often be unrealistic and extremely expensive [12].

1. **Dynamic Pricing:** Controlling supply and demand in real time by changing the price has a myriad of problems associated. It focuses on local short term optimization and raises questions regarding economic fairness to its user groups [13].
2. **Distributed Control and AIMD-Inspired Algorithms:** These use localized decisions for optimization, focusing on scalability and privacy. Additive Increase / Multiplicative Decrease (AIMD), widely used in TCP network control, allocates resources based on a global capacity aiming for long-term optimized contracts [14] [15].

Even sophisticated distributed algorithms often focus on converging to optimal rates, not full probabilistic allocation plans for any state. Inefficient allocation of resource have far-reaching consequences, well beyond economic impacts to the designers.

2.3 Fairness and Quality of Service (QoS)

Beyond efficiency, SE schemes' viability depends on serving correlated groups satisfactorily. SE fairness includes equitable access (demand groups), fair income (providers), and avoiding biases. The dissertation proposes a plan to let scheme designers get a stronger view on understanding the relationship between competing demands and hybrid supplies to make informed changes. The platform's public perception relies heavily on the ability to guarantee a certain Quality of Service (QoS) [16]. This primarily has concerns relating to waiting times, availability, and rejection to resource [14]. A key challenge lies to move beyond a deterministic view or a quantile based QoS metric towards probabilistic guarantees, e.g. ensuring that the probability of undersupply is within some reasonable limit.

2.4 Proposed Approach: FPD-OT for Resource Redistribution in SEs

This dissertation claims that a significant gap exists in the literature for a framework capable of handling multivariate SE systems while providing probabilistic, cost-optimal, and QoS-aware allocation policies. It is proposed to fill this gap by leveraging Fully Probabilistic Design (FPD), using entropy-regularized Optimal Transport (OT) as the key tool. FPD is a Bayesian decision making approach to design probability distributions or *optimal plans* under some knowledge constraints whilst minimizing a cost. The key advantage of FPD-OT is to get joint probability distributions as which can be treated as interpretable plans providing the scheme designer with a multitude of tuning knobs and actionable insights. The dissertation further develops the methodologies of the application by introducing additional linear constraints to incorporate notions of QoS and propose a modified algorithm to find optimal solutions.

3 Theoretical Framework: FPD-OT

The following chapter lays the foundational theories for the proposed resource redistribution framework. Optimal Transport (OT) and Fully Probabilistic Design (FPD) principles are introduced.

3.1 Introduction to Optimal Transport (OT)

An early version proposed by Gaspard Monge in the 18th century [17], OT is a branch of mathematics allowing for the most efficient way to transform one probability distribution into another, given a certain cost. Monge's aim was to map piles of sand onto different locations to build fortifications, with the minimum moving cost. Since then, the problem has been refined and adapted to modern applications such as computer graphics and vision [18] [19], machine learning for generative models [20], and even AI data repair [21]. Let

$$\Omega_S = \{0, 1, \dots, Q_{\max}\}, \quad \Omega_D = \{0, 1, \dots, Q_{\max}\},$$

and let $\mu(\mathbf{x})$ and $\nu(\mathbf{y})$ be the supply and demand PMFs on these sets (see Table 1). We seek a joint plan $\pi(x, y)$ whose marginals recover μ and ν , minimizing the expected cost of matching supply x to demand y .

3.1.1 Monge vs Kantorovich Formulations

The two primary formulations of the OT problem are:

1. **Monge's Formulation:** This method concerns itself with finding a deterministic (one to one) **transport map** $T : \Omega_S \rightarrow \Omega_D$, assigning each point x in the source domain Ω_S to a unique point y in the target domain Ω_D . The Monge map is restrictive as it does not allow for splitting the mass at x . Thus for discrete distribution, a solution might not always have a feasible Monge OT map [22].
2. **Kantorovich's Relaxation:** In the 1940s, Leonid Kantorovich [23] proposed a more general formulation which sought to make a probabilistic **transport plan** (instead of a mapping), denoted as $\pi(x, y)$. This plan is a joint probability distribution on $\Omega_S \times \Omega_D$ and its marginals are constrained to be $\mu(\mathbf{x})$ and $\nu(\mathbf{y})$. We aim to find the optimal joint probability distribution, $\pi^O(\mathbf{x}, \mathbf{y})$ by minimizing a certain cost, $C(\mathbf{x}, \mathbf{y})$. Thus:

$$\pi^O(\mathbf{x}, \mathbf{y}) = \min_{\pi \in \Pi(\mu, \nu)} \sum_{x \in \Omega_S} \sum_{y \in \Omega_D} C(\mathbf{x}, \mathbf{y}) \pi(\mathbf{x}, \mathbf{y}) \quad (1)$$

where $\Pi(\mu, \nu)$ is the set of all joint plans π on $\Omega_S \times \Omega_D$ such that the following **marginal constraints** are met:

$$\sum_{y \in \Omega_D} \pi(x, y) = \mu(\mathbf{x}) \quad \forall x \in \Omega_S \quad (2)$$

$$\sum_{x \in \Omega_S} \pi(\mathbf{x}, \mathbf{y}) = \nu(\mathbf{y}) \quad \forall y \in \Omega_D \quad (3)$$

This relaxation allows fractionally transporting points from X to points in Y . The $n \times m$ transport matrix, π , forms a probability simplex Δ^{nm-1} . This serves as linear constraints to the OT problem ensuring non-negativity and normalization.

$$\Delta^{nm-1} = \left\{ \pi \in \mathbb{R}^{n \times m} \mid \pi_{ij} \geq 0 \quad \forall i, j \text{ and } \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} = 1 \right\} \quad (4)$$

This can thus be represented as:

$$\pi \in \Pi(\mu, \nu) = \left\{ \pi \in \Delta^{nm-1} \mid \mu = \pi \mathbf{1}_m, \nu = \pi^\top \mathbf{1}_n \right\} \quad (5)$$

3.1.2 Core SE Context in OT

The key OT concepts with SE contextualizations are the following:

1. **Marginals** (μ, ν) : In the SE scenarios, these are the distributions we are trying to match, the historical or forecasted probabilities of available supply and demand, respectively. For a simple 1D scenario, $\mu(\mathbf{x})$ is a Probability Mass Function (PMF) representing the probability that the available supply is exactly x units, similarly for demand and $\nu(\mathbf{y})$. This dissertation concerns itself with discrete quantities ranging from 0 to a maximum realistic capacity of Q_{max} . Thus, $x \in \Omega_S = \{0, 1, \dots, Q_{max}\}$ and $y \in \Omega_D = \{0, 1, \dots, Q_{max}\}$. A visualization can be seen in Figure 1 where the distributions represent a univariate supply and demand. The dissertation will eventually use multivariate marginals to extend the PMFs to $\mu(x_1, x_2)$ to represent two distinct but correlated supply sources, and similarly for demand.
2. **Cost Metric** $(C(\mathbf{x}, \mathbf{y}))$: This is the function we aim to minimize when matching the supply and demand distributions. For the research, this cost looks purely at the economic operational costs of running SE schemes incurred by the scheme designer. The costs might be related to under-supplying, over-supplying or having a supply beyond a certain threshold.

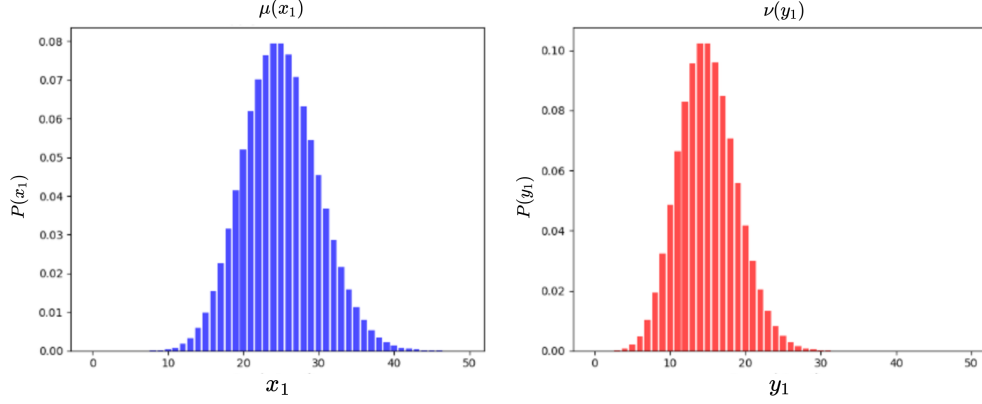


Figure 1: Example Supply ($\mu(\mathbf{x})$) and Demand ($\nu(\mathbf{y})$) Probability Mass Functions

3. **Joint Optimal Plan ($\pi(x, y)$):** The key output for an OT problem, the joint probability distribution represents the optimal probability that in a case with x available supply should be matched with a state of y demanded units. Rather than matching, it is thought of a probabilistic allocation policy. For a 1D supply and demand, the plan is represented in 2D but will increase in dimensions for multivariate cases.

Extensive research has been done to solve the OT problem with minimal computational cost. The classical Kantorovich OT problem is a linear program but can be demanding to solve for large state spaces. Cuturi et. al introduced entropy regularization to the OT problem by adding a penalty term to the objective function [22].

$$\pi^O(\mathbf{x}, \mathbf{y}) = \min_{\pi \in \Pi(\mu, \nu)} \sum_{x \in \Omega_S} \sum_{y \in \Omega_D} C(\mathbf{x}, \mathbf{y}) \pi(\mathbf{x}, \mathbf{y}) + \epsilon KL(\pi || \phi) \quad (6)$$

where $KL(\pi || \phi)$ is the Kullback-Leibler divergence from the distribution $\pi(x, y)$ to a base distribution $\phi(x, y)$. ϵ is the regularization constant. The KL divergence is calculated by [24]:

$$KL(\pi || \phi) = \sum_{x \in \Omega_S} \sum_{y \in \Omega_D} \pi(\mathbf{x}, \mathbf{y}) \log\left(\frac{\pi(\mathbf{x}, \mathbf{y})}{\phi(\mathbf{x}, \mathbf{y})}\right) \quad (7)$$

With the entropic term, the problem becomes **strictly convex** and more importantly allows the use of **Sinkhorn-Knopp (SK) algorithm** to approximate the joint probability distribution. The SK algorithm is an iterative scaling method which is computationally efficient and allows the satisfaction of the marginal constraint whilst minimizing KL divergence. The dissertation explains the modifications to the SK algorithm in later sections. [22]

3.2 Introduction to Fully Probabilistic Design (FPD)

Rooted in the Bayesian Decision Theory [25], the central aim is to determine an optimal PMF, $\pi^O(\mathbf{x}, \mathbf{y})$ which is “as close as possible” to a predefined **Ideal Distribution**, ($\pi^I(\mathbf{x}, \mathbf{y})$), whilst maintaining a set of **Knowledge Constraints** [26].

3.2.1 Core FPD Concepts and SE Context

1. **Ideal Distribution ($\pi^I(\mathbf{x}, \mathbf{y})$):** This is the joint plan that the scheme designer would choose if there were no real life constraints. It embodies the preferences and economic objectives of

the SE scheme. It is derived from three key components:

- (a) The **Cost Metric** ($C(\mathbf{x}, \mathbf{y})$) which reflects the cost of every supply-demand contract (or pairing).
- (b) The **Regularization Parameter** (ϵ) controls the sensitivity of the cost.
- (c) A **Base Distribution** ($\phi(\mathbf{x}, \mathbf{y})$) encodes any prior beliefs one might have about the joint probability distribution. If none, it is usually taken to be a uniform distribution.

The Ideal Distribution takes the form of a Boltzmann Distribution where,

$$\pi^I(\mathbf{x}, \mathbf{y}) \propto \phi(\mathbf{x}, \mathbf{y}) e^{-C(\mathbf{x}, \mathbf{y})/\epsilon} \quad (8)$$

The distribution has interesting parallels with thermodynamics [27] where the Ideal Distribution is a probability distribution of a system given its energy (Cost Metric) and temperature (ϵ).

2. **Knowledge Constraints:** These are hard constraints that have to be necessarily met by the optimal distribution $\pi^O(\mathbf{x}, \mathbf{y})$. For SE applications, these constraints are initially just the observed marginal PMFs for supply and demand. This ensures that, in the long run, the decisions by the policy $\pi^O(\mathbf{x}, \mathbf{y})$ will adhere to the historical marginals.

3.2.2 Mathematical Formulation

The “closeness” of the ideal distribution and the optimal distribution is measured again using the KL divergence. Thus, the mathematical formulation becomes:

$$\pi^O(\mathbf{x}, \mathbf{y}) = \arg \min_{\pi \in \Pi(\mu, \nu)} KL(\pi || \pi^I) \quad (9)$$

3.3 Regularized OT as a Special Case of FPD

Boufelja et al. formalized the correlation between FPD and Entropic Regularized OT [28]. It allows extending OT using FPD’s mechanisms for handling more sophisticated knowledge constraints and uncertainty. When the Ideal Distribution takes the form of the Boltzmann Distribution, the same Cost Function and Regularisation parameter are used as in Regularized OT. The Joint Plan $\pi^O(\mathbf{x}, \mathbf{y})$ obtained by solving the entropy-regularized OT problem is identical to the optimal FPD plan found by minimizing $KL(\pi || \pi^I)$ subject to the marginal constraints. The knowledge constraints in the FPD program are the marginal constraints mentioned in OT. Combining the two tools allows us to eventually add Quality of Service constraints to cater the solution to SE contexts.

3.4 FPD-OT in SE Scheme Design

There has only been a handful of intersection between FPD, OT and SE Scheme Design. OT research in resource allocation has been over networks due to the real-world applicability. Hughes and Chen [29] research recognizes the tradeoff between efficiency and fairness of OT Plans. Although the paper limits itself to networks, ideas from it can be taken forward in this project specifically. Another use of OT in networks was seen in Hughes and Chen [29] where path planning was the key motive. Corless et al. [30] utilized the concept of constrained OT where some target points may be prohibited from certain sources. The authors proposed a modified version of the Sinkhorn-Knopp algorithm. With the example of EV charging in a decentralized grid, the research allows for a more practical view of supply and demand.

The well-known book *Optimal Transport Methods in Economics* connects OT with relevant theories from economics giving interesting approaches to traditional problems such as pricing and market equilibrium. Though it provides a comprehensive list of applications, it lacks specific examples of sharing economies [31].

The previous chapter mentioned the lack of distributional knowledge used in deterministic and single value probabilistic methods [12]. The paper proposes FPD-OT as a superior alternative to make allocation decisions in SE schemes, offering advantages mentioned in Table 2:

Table 2: Advantages of FPD-OT for Allocation Decisions in SE Schemes

Advantage	Description
Intrinsic Probabilistic Nature	As mentioned, supply and demands are stochastic processes. More importantly, these distributions can have multimodal natures (representing surges) which is reflected in the joint probabilistic plans provided by FPD-OT.
Cost and Constraint Compromises	FPD-OT allows us to combine economic preferences (Cost Function) whilst making sure empirical accuracies (Marginals) remain.
Actionable Insights	The Joint Plans can be used in various ways, including conditionals and marginals in order to get specific insights instead of it being a black box.
QoS Extension	The FPD-OT formulation allows us to add linear constraints beyond the marginals, which can be represented as guaranteed service levels in SE schemes.

4 Three Scenarios of SE Schemes

To explore FPD-OT framework’s capabilities in real-world SE schemes, the dissertation develops and analyses models for schemes with increasing complexity.

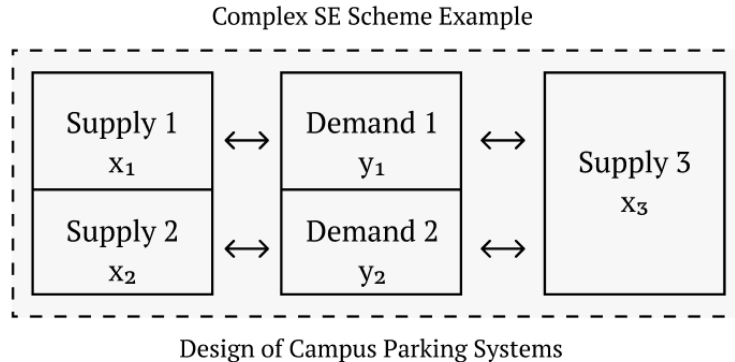


Figure 2: Conceptual parking-allocation example from [8], illustrating an SE scheme with hybrid supply (university-owned vs. landlord spaces) and competing demand (staff vs. resident parkers)

A campus parking system example [8] illustrates such complexity (Figure 2), featuring hybrid supplies (university and residential spaces) and multivariate demand (university community, land-

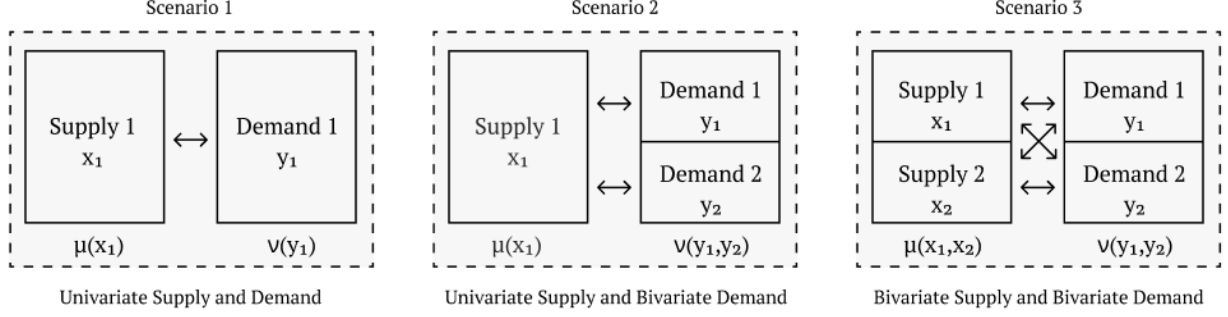


Figure 3: Schematic representation of the three SE scenarios of increasing complexity analyzed in this dissertation: (Left) Scenario 1: Univariate supply $\mu(x_1)$ and univariate demand $\nu(y_1)$. (Center) Scenario 2: Univariate supply $\mu(x_1)$ and bivariate (competing) demands $\nu(y_1, y_2)$. (Right) Scenario 3: Bivariate (hybrid) supply $\mu(x_1, x_2)$ and bivariate (competing) demands $\nu(y_1, y_2)$

lords) to address demand surges. To assure a QoS to participating landlords, the university parking was further divided into Reserved and Unreserved spaces. The authors were primarily concerned with single-value probabilistic strategies for allocation. Motivated by the parking design, this dissertation will incrementally analyse three stages, seen in Figure 3 of SE resource allocation problems using FPD-OT:

1. **Single Supply and Single Demand:** Establishing the simplest FPD-OT model, introducing the core mechanics, cost structures, and interpretative tools for matching single supply and demand marginals.
2. **Single Supply with Competing Demands:** Extends the model to allocate a single supply to multiple correlated demand segments. This will involve a 2D demand marginal.
3. **Hybrid Supply with Competing Demands:** Addresses the most complex scenario with multiple supply sources and demand groups, involving 2D supply and demand marginals, yielding a 4D FPD-OT plan.

4.1 Implementation Strategy

The above-mentioned FPD-OT problems in this dissertation are solved computationally (Link to code in Appendix B). Initially the Python OT library was explored in order to generate the plans, but it lacked capabilities of incorporating Probabilistic knowledge (crucial for FPD-OT) and thus the codebase was adapted and customised for the specific sharing economy scenarios. The core of the methodology relies on the Sinkhorn-Knopp algorithm which minimized $KL(\pi || \pi^I)$ subject to the marginal constraints.

The marginal supply ($\mu(\mathbf{x}, \mathbf{y})$) and demand ($\nu(\mathbf{x}, \mathbf{y})$) distributions were modelled using truncated bimodal Poisson distributions. The multimodality allows us to explore demand surges and supply shocks with certain given probabilities. The truncation (e.g. between 0 and Q_{max}) enforces operational limits or realistic scenarios.

For all the FPD-OT scenarios, the base distribution, ϕ is assumed to be uniform and ϵ is kept at 0.1.

4.2 Scenario 1: Univariate Supply and Univariate Demand

This section introduces the simplest but an illustrative case of matching a single stochastic supply source to a demand distribution. While arbitrary, it allows us to establish the core ideas of the methodology before coming to complicated scenarios.

4.2.1 Supply and Demand Marginals: $\mu(x_1), \nu(y_1)$

The marginals are built upon historical data from two different scenarios which require coupling. For example, in a car sharing scheme, the supply marginal might be based on Location A, whilst the demand marginal is based on Location B. The scheme designer is looking to start the SE instance in location C and thus needs a coupling between the two. The marginal distributions are defined on these sets:

- **Supply Marginal Distribution $\mu(x_1)$:** In this discrete PMF, $\mu(x_1)$ represents the probability that the available supply is exactly x_1 units. It represents the probability of the number of cars being available at a given timestamp by the prosumers.
- **Demand Marginal Distribution $\nu(y_1)$:** Similarly, $\nu(y_1)$ represents the probability that the required demand is exactly y_1 units. It is built by collecting historical data of demand of the resource. This would be request patterns of shared resources over a long period of time.

To model these marginal distributions, this dissertation will employ mixtures of Poisson distributions. The fundamental Poisson PMF is given by:

$$P(K = k|\lambda) = \frac{e^{-\lambda} \lambda^k}{k!} \quad \text{for } k = 0, 1, 2, \dots$$

where λ is the average quantity of a resource, and k is the specific supply or demand.

To construct our bimodal marginals, we first define an unnormalized mixture of two Poisson distributions. For the supply marginal $\mu(x)$, let $\mu'(x)$ be this unnormalized mixture:

$$\mu'(x|\lambda_{s1a}, \lambda_{s1b}, p_s) = p_s \cdot \frac{e^{-\lambda_{s1a}} \lambda_{s1a}^x}{x!} + (1 - p_s) \cdot \frac{e^{-\lambda_{s1b}} \lambda_{s1b}^x}{x!}$$

where:

- λ_{s1a} and λ_{s1b} are the means of the two underlying Poisson components for supply.
- p_s is the mixture weight for the first component. It represents the probability of the first state of the distribution happening in real life. ($0 \leq p_s \leq 1$).

To ensure $\mu(x)$ is a valid PMF over the discrete and finite set $\Omega_S = \{0, 1, \dots, Q_{\max}\}$, it must be truncated and normalized $\mu'(x)$:

1. **Truncation:** Consider $\mu'(x)$ only for $x \in \Omega_S$.
2. **Normalization:** The final supply marginal $\mu(x)$ is obtained by:

$$\mu(x) = \frac{\mu'(x|\lambda_{s1a}, \lambda_{s1b}, p_s)}{\sum_{k=0}^{Q_{\max}} \mu'(k|\lambda_{s1a}, \lambda_{s1b}, p_s)} \quad \text{for } x \in \Omega_S$$

and $\mu(x) = 0$ for $x \notin \Omega_S$.

The demand marginal is created in a similar way. For the example, we assume the following values:

$$\lambda_{s1a} = 15 \quad \lambda_{s1b} = 30 \quad p_s = 0.4$$

$$\lambda_{d1a} = 20 \quad \lambda_{d1b} = 40 \quad p_d = 0.7$$

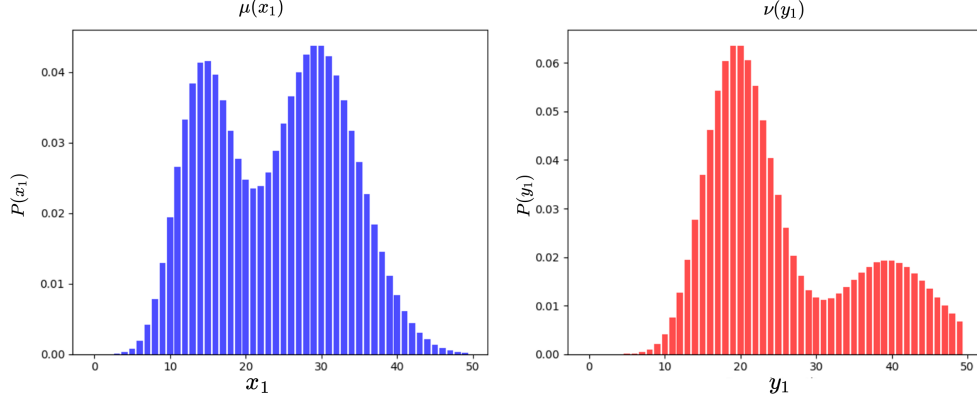


Figure 4: Bimodal marginal Probability Mass Functions (PMFs) for Scenario 1: (Left) Univariate supply distribution $\mu(x_1)$ with modes at approximately 15 and 30 units. (Right) Univariate demand distribution $\nu(y_1)$ with modes primarily around 20 units and a smaller mode near 40 units. These distributions, based on parameters specified in 4.2.1, represent the stochastic nature of supply and demand, including potential surges and spikes.

4.2.2 Base Distribution ($\phi(x_1, y_1)$) and Regularization Constant (ϵ)

$\phi(x, y)$ is assumed to be the uniform distribution. The regularization constant, ϵ determines the smoothness of the joint FPD-OT Plan, controlling the influence of the cost metric.

4.2.3 Designing a Cost Metric for SE Schemes: $C(x_1, y_1)$

The ideal distribution relies on the cost function $C(x_1, y_1)$ which is the cost incurred when x_1 units are supplied with a demand of y_2 units. The goal of the research is to give the scheme designer an open-hand to design case-specific cost metrics. For this exploration, a generally applicable cost function was designed as a linear combination of multiple operational scenarios. As the cost metric is represents the “energy” of the system, it is taken into consideration that the lowest cost is 0. The cost is defined as:

$$C(x_1, y_1) = c_o \max(0, x_1 - y_1) + c_u \max(0, y_1 - x_1) + c_t \max(0, x_1 - x_{thresh})$$

The first two components are derived from the classic “Newsvendor Problem” where the supplier incurs asymmetric costs by over and under-provisioning. The breakdown can be seen in Table 4.2.3. With the example parameters chosen to be $c_u = 10, c_o = 1$ and $c_t = 5$ with a threshold at $x_{thresh} = 20$, Figure 5 shows the visualization of the cost metric as a 2D tensor. The cost is asymmetric along the line where supply equals demand. It increases much more quickly when supply falls short of demand, compared to the opposite scenario. Further, the thresholding penalty is also seen that up until $x_1 = 30$ the cost remains low but starts to increase beyond that. The

Table 3: Decomposition of the single-resource cost function $C(x_1, y_1)$

Component	Condition	Cost term	Economic interpretation
Oversupply	$x_1 > y_1$	$c_o \max(0, x_1 - y_1)$	Inventory holding, spoilage, obsolescence, or disposal costs arising from excess supply.
Undersupply	$y_1 > x_1$	$c_u \max(0, y_1 - x_1)$	Lost sales, platform-reputation damage, and network-effect leakage when demand is unmet; empirical studies often find $c_u > c_o$ [32].
Threshold exceedance	$x_1 > x_{\text{thresh}}$	$c_t \max(0, x_1 - x_{\text{thresh}})$	Penalties for breaching storage limits, regulatory caps, or operational complexity beyond a safe capacity.

Ideal Plan, mentioned in Equation 8 is a direct result of the manipulation of the costing parameters and can be seen in Figure 5.

4.2.4 Mathematical Formulation

The aim is to find the optimal transport plan $\pi^O(x_1, y_1)$ such that

$$\pi^O(x_1, y_1) = \underset{\pi \in \Pi(\mu, \nu)}{\operatorname{argmin}} KL(\pi || \pi^I)$$

subject to

$$\sum_{y_1} \pi(x_1, y_1) = \mu(x_1) \quad \forall x_1; \quad \sum_{x_1} \pi(x_1, y_1) = \nu(y_1) \quad \forall y_1$$

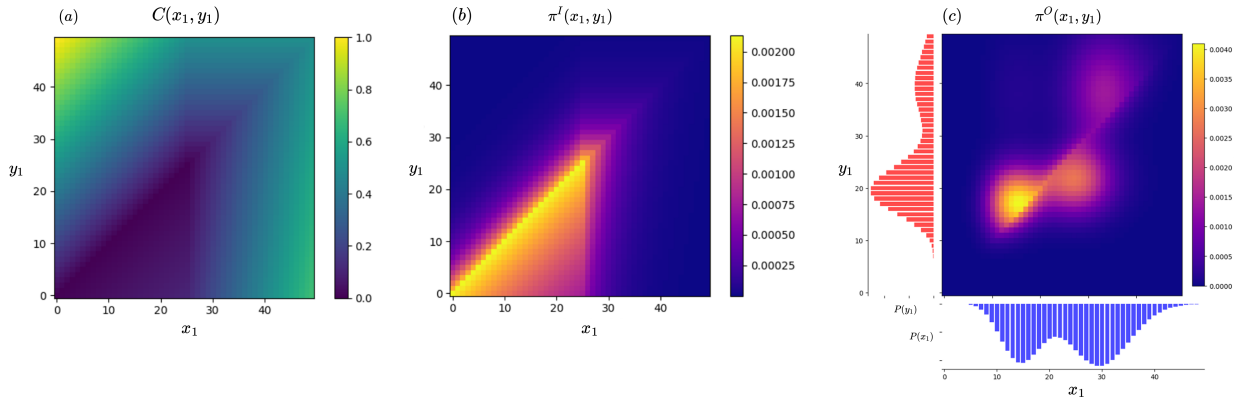


Figure 5: Key Components of the FPD-OT Framework for Scenario 1: (a) Cost Matrix $C(x_1, y_1)$ illustrating rising penalties for oversupply, undersupply and for exceeding a supply threshold. (b) Ideal Plan $\pi^I(x_1, y_1)$ derived from the cost matrix, regularization parameter and the base distribution. It doesn't satisfy the marginals and thus lies outside the feasible set $\Pi(x_1, y_1)$. (c) optimal joint plan, $\pi^O(x_1, y_1)$, satisfying the input marginals $\mu(x_1)$ (bottom plot) and $\nu(y_1)$ (left plot), representing the cost-optimal allocation policy.

4.2.5 Optimal Joint Plan: $\pi^O(x_1, y_1)$

The heatmap in Figure 5 shows the optimal joint plan, $\pi^O(x_1, y_1)$ conveying the optimal probability of matching x_1 units of supply to y_1 units of demand. A sanity check was performed to confirm if the marginal constraints were met. It can be seen that the hotspots on the joint plan is influenced by the bimodal nature of the marginals where a high probability exists at the intersection of the modes. Even though we have a high weighting on the undersupply component of the cost function, the plan needs to satisfy the marginals and thus still inherently has a probability of around 0.54 of undersupplying. Designers extract the insight that undersupplying might be unavoidable, but the FPD-OT plan minimizes as much as possible. This is visible as most of the mass is along the line where supply meets demand and the plan doesn't deviate much from there. To reiterate, this generated plan is the closest policy to the Ideal Distribution seen in Figure 5. With the optimal plan, the dissertation will aim to apply **inference operations** such as conditionals and induced marginals to get a deeper understanding about the SE schemes.

4.2.6 Strategic Insights from Conditionals

Conditional probabilities are a key tool for generating actionable intelligence from the FPD-OT policy. It analyses specific slices of the optimal plan to derive strategies.

- **Optimal Supply Given Demand $\pi^O(x_1|y_1)$:** Figure 6 shows how the FPD-OT plan looks like when the demand is realized to be five different scenarios. The histograms are a result of slicing the joint plan at the realization. The histograms capture the bimodal nature of the distribution and show that even though at a certain time, the demand forecast might be e.g. 25, the optimal probability of supply (based on historic marginals) has a wide spread. The scheme designer has multiple actions they can choose from, such as picking the expected value of the distribution, the maximum a posteriori (MAP) or even simply sampling from the distribution, to decide on the supply. This helps in short term incentives to peer suppliers to increase or decrease supply based on the demand realisations.
- **Optimal Demand Given Supply $\pi^O(y_1|x_1)$:** Similarly, Figure 7 shows the optimal demand distributions if the supply is abruptly constrained to specific values (e.g. provider's logging off, fleet maintenance.).

4.3 Scenario 2: Univariate Supply and Bivariate Demands

Building upon the previous section, we can now extend the FPD-OT framework to tackle a more complex narrative that is often present in sharing economies: matching a single source to two distinct demands. This is often seen in cases where there are two demographics that need to be served or two different locations but the source pool is still singular. For instance, in a ride-sharing context, "economy" riders and "business class" riders both require vehicles, but the latter might have a higher willingness to pay and a lower tolerance for delays.

4.3.1 Supply and Demand Marginals: $\mu(x_1)$, $\nu(y_1, y_2)$

The supply side remains characterized by a univariate bimodal marginal probability distribution $\mu(x_1)$ over the discrete set $\Omega_S = \{0, 1, \dots, Q_{\max}\}$.

The change lies in the demand characterization. We now model demand as a bivariate quantity $\nu(y_1, y_2)$, where y_1 and y_2 represent the number of units requested by the two distinct demand

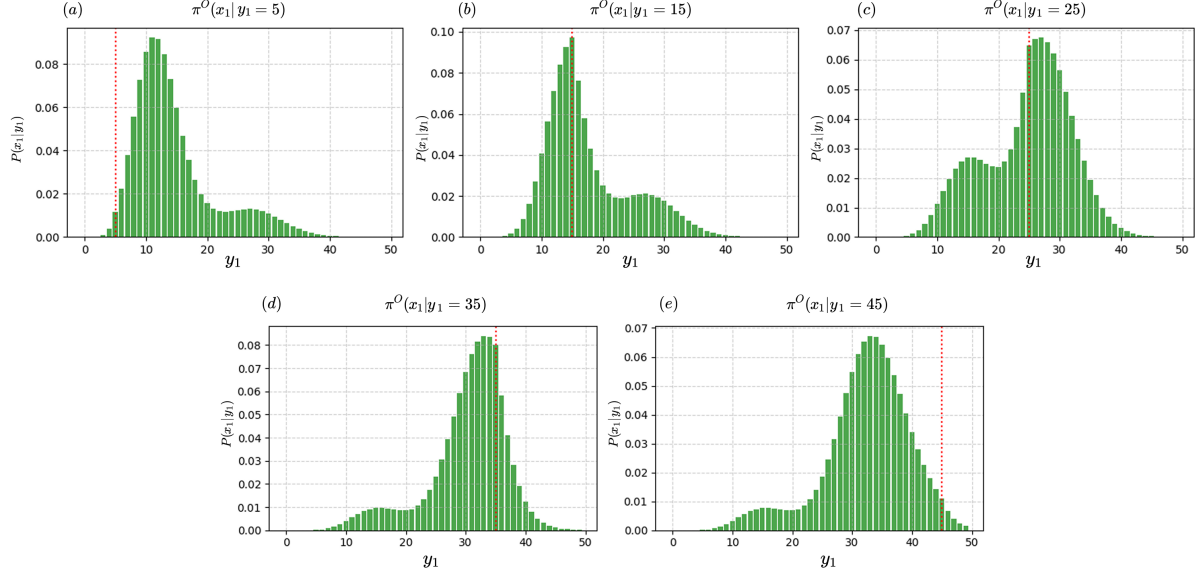


Figure 6: Conditional probability distributions of optimal supply, $\pi^O(x_1|y_1)$, for Scenario 1, given five different realizations of demand $y_1 \in \{5, 15, 25, 35, 45\}$. Each histogram illustrates the probabilistic guidance for supply allocation when a specific demand level is observed.

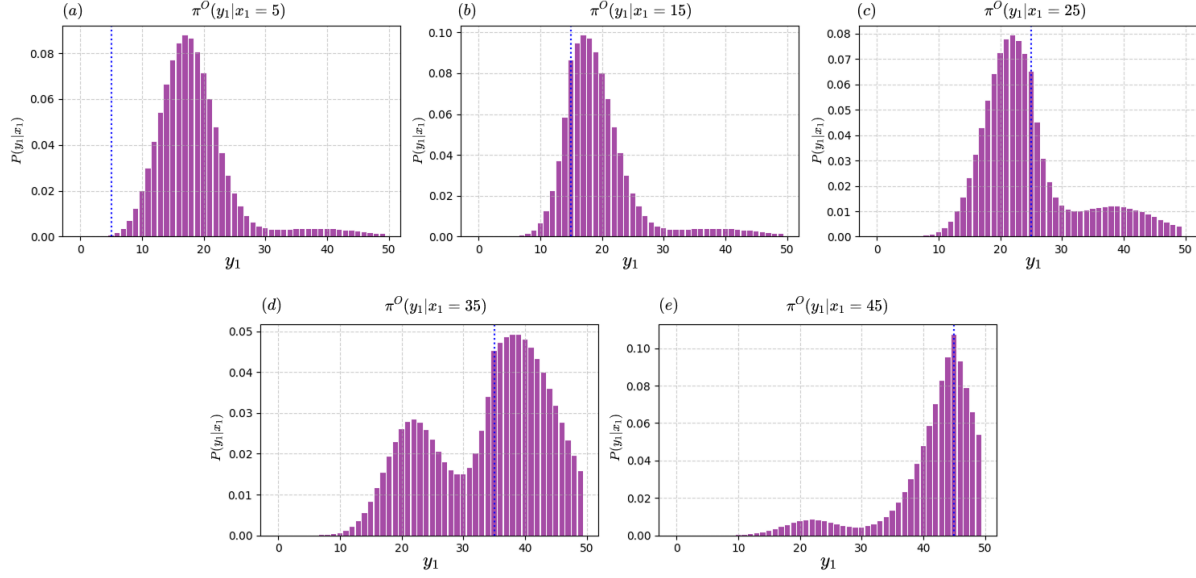


Figure 7: Conditional probability distributions of optimal demand, $\pi^O(y_1|x_1)$, for Scenario 1, given five different realizations of supply $x_1 \in \{5, 15, 25, 35, 45\}$. Each histogram shows the likely demand distribution the system should prepare for if supply is constrained to a specific level.

sources. This distribution captures not only the individual demand patterns but also the correlation between them. There might be cases where an increase in demand happens simultaneously for both groups. We use a 2D truncated bimodal bivariate poisson for the numerical analysis. Detailed

derivation of bimodal bivariate Poissons in Appendix A.

The unimodal bivariate Poisson is created by assuming 2 independent truncated Poisson distributions (with means $\lambda_{d1a}, \lambda_{d2a}$ and then adding a covariance factor (λ_{d12a}) signalling their dependence. The multimodality was achieved by using a similar idea with weights, as previously mentioned. A detailed description can be found in the appendix A.

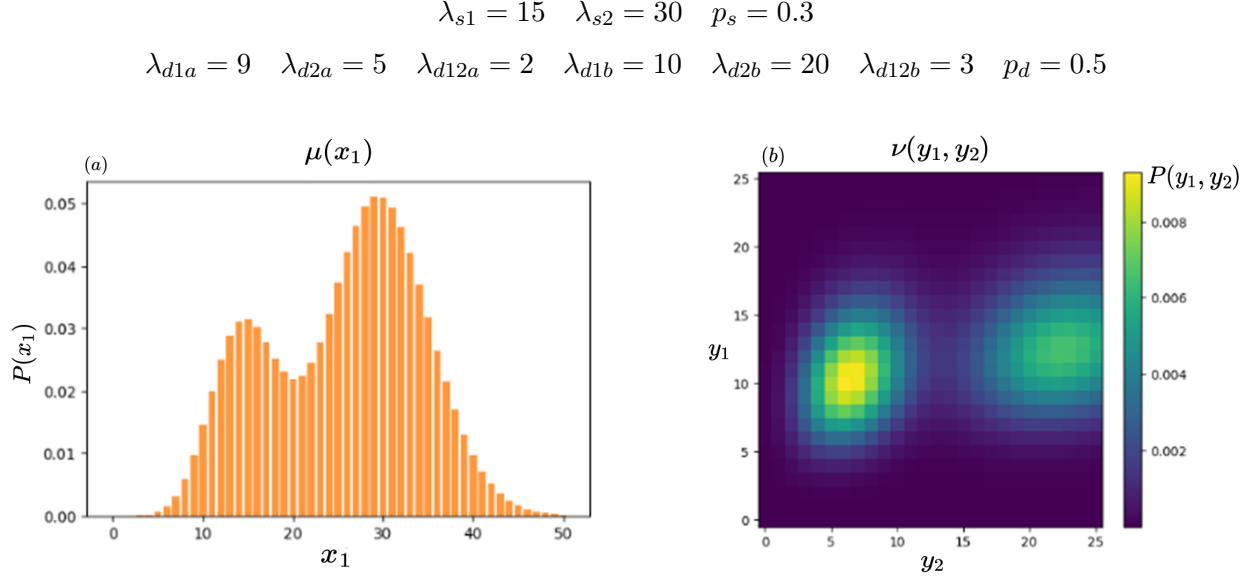


Figure 8: Input marginal probability distributions for Scenario 2: (a) Univariate bimodal supply distribution ($\mu(x_1)$). (b) Bivariate bimodal demand distribution ($\nu(y_1, y_2)$), capturing correlated demands from two distinct groups.

The feasible set $\Pi(\mu, \nu)$ now consists of transport plans $\pi(x_1, y_1, y_2)$ defined over the 3D space $\Omega_S \times \Omega_D$. A valid plan π must satisfy the marginal constraints:

$$\sum_{y_1=0}^{Q_{\max}/2} \sum_{y_2=0}^{Q_{\max}/2} \pi(x_1, y_1, y_2) = \mu(x_1), \quad \forall x \in \Omega_S$$

$$\sum_{x_1=0}^{Q_{\max}} \pi(x_1, y_1, y_2) = \nu(y_1, y_2), \quad \forall (y_1, y_2) \in \Omega_D$$

4.3.2 Adaptation of the Cost Metric $C(x_1, y_1, y_2)$

The cost function $C(x_1, y_1, y_2)$ needs to be adapted to reflect the cost incurred when supply x_1 meets the combined demand pressure from y_1 and y_2 . We modify the cost metric such that:

$$C(x_1, y_1, y_2) = c_o \max(0, x_1 - (y_1 + y_2)) + c_u \max(0, (y_1 + y_2) - x_1) + c_t \max(0, x_1 - x_t)$$

While this specific formulation does not differentiate the undersupply costs for y_1 and y_2 individually, the framework allows for more granular cost components. For instance, terms like $c'_{u1} \cdot \max(0, y_1 - x_1^*)$ and $c'_{u2} \cdot \max(0, y_2 - x_1^*)$ could be incorporated to target specific pairings. The design of $C(x, y_1, y_2)$ is critical as it encodes the SE platform's objectives regarding efficiency, customer satisfaction for different segments, and operational constraints. For the numerical example,

the following parameters were chosen: $c_u = 10$, $c_o = 2$ and $c_t = 5$ with a threshold at $x_{thresh} = 20$. With a 2D demand and a 1D supply, the cost metric becomes a 3D tensor with each coordinate representing the energy of that specific combination of supply and demands. Figure 9 shows multiple slices of the cost metric, conditioned at different supply values. The solid 3D tensor would have been tough to visualize.

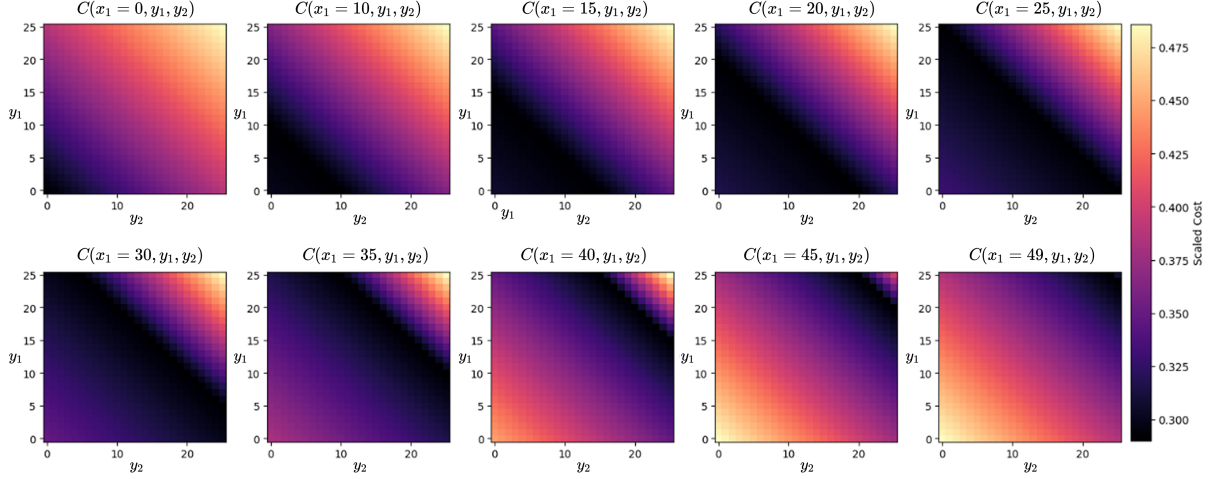


Figure 9: Cost matrix heatmap at different values of x_1 . Visualizations of the cost metric $C(x_1, y_1, y_2)$ for Scenario 2. Each heatmap represents a 2D slice of the 3D cost tensor for a fixed supply level x_1^* , showing the cost of allocating units to various combinations of demand y_1, y_2

4.3.3 Optimal Joint Plan for Competing Demands $\pi^O(x_1, y_1, y_2)$

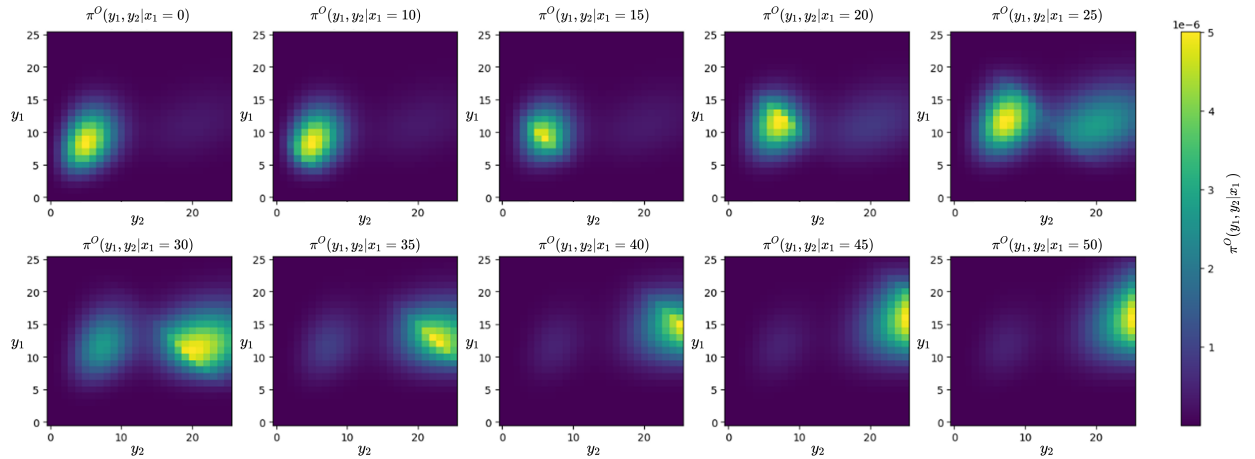


Figure 10: Slices of the optimal joint plan for Scenario 2, showing the conditional probability distribution of bivariate demands $\pi^O(y_1, y_2 | x_1^*)$ given specific realizations of supply x_1^* . Each heatmap indicates the most probable demand pair y_1, y_2 to be served by the available supply x_1^*

Similar to the cost sensor, the optimal joint plan is a 3D tensor. We analyse 2D slices conditioned on fixed supply levels x_1^* . For a given x_1^* , the hotspots (high probability regions) indicate the most

likely combinations of (y_1, y_2) that this supply level will serve. With limited resources, $x_1 < 15$, the platform’s optimal strategy is to primarily serve the lower-demand user segment (Mode 1). Attempting to serve the higher-demand mode would lead to severe undersupply, making it an unattractive option according to the plan.

Conditional Supply Distributions ($\pi^O(x_1 | y_1, y_2)$): Figure 11 shows five distinct plots. Each histogram shows, for a given random demand pair (y_1, y_2) , the probability that the optimal plan allocates a supply level x_1 . This is the most operationally relevant visualization. When the platform observes a specific demand (y_1, y_2) , this conditional distribution provides direct guidance for resource allocation. The scheme designer can use this information in several ways:

- **Operational Strategy Selection:** The shape of the conditional distribution $\pi^O(x_1 | y_1, y_2)$ informs different operational choices. As mentioned before, the SE scheme designer knows the shape of the probabilistic supply which would be optimal for the demands, and can introduce necessary incentives.
- **Stress Testing and Scenario Planning:** By examining the conditional supply distributions $\pi^O(x_1 | y_1, y_2)$ for particularly challenging demand pairs (e.g., very high total demand $y_1 + y_2$, or scenarios known to strain resources), the designer can understand the system’s optimally planned response under pressure. This helps in identifying potential bottlenecks or evaluating the robustness of the supply chain.

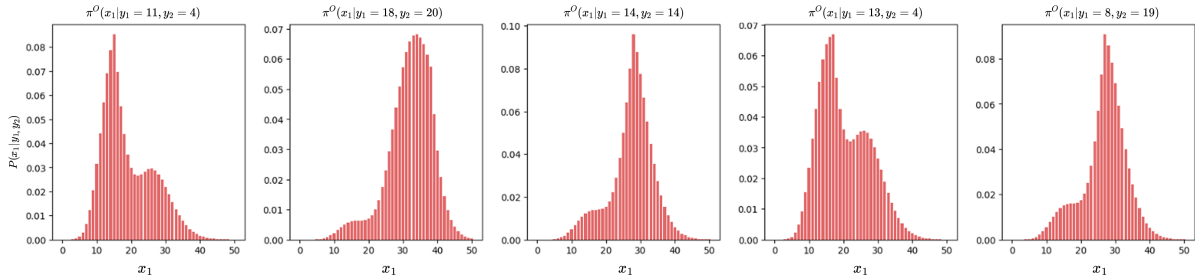


Figure 11: Conditional probability distributions of optimal supply $\pi^O(x_1 | y_1^*, y_2^*)$ for Scenario 2, given five distinct realizations of bivariate demand pairs y_1^*, y_2^* . Each histogram provides actionable guidance on the optimal supply x_1 to allocate when a specific demand combination from two competing groups is observed.

Together, these visualizations provide a comprehensive picture of the FPD-OT model’s application, moving from the input assumptions (marginals) and economic drivers (costs) to the resulting optimal plan and its actionable implications for managing resources in a sharing economy with competing demands.

4.3.4 Further Sharing Economy Considerations

The total expected cost of the optimal plan, $E_{\pi^O}[C(x_1, y_1, y_2)] = \sum_{x_1, y_1, y_2} \pi^O(x_1, y_1, y_2) C(x_1, y_1, y_2)$, is an important performance indicator. It is the minimal achievable operational cost given the input distributions and cost metric. The FPD-OT model enables sensitivity analysis by varying cost parameters ($c_o, c_u, c_t, x_{\text{thresh}}$) or the regulariser (ϵ). Understanding their impact on the optimal plan $\pi^O(x_1, y_1, y_2)$ and total expected cost informs policy setting. Figure 10 shows a mode switching mechanism. With supply increases, the joint plan probabilistically allocates from one mode to

another. When $x_1 = 30$, the policy might lean towards the mode at $\{y_1, y_2 = 10, 20\}$ but shows that there are probabilities of serving the lower mode at the same time. At those points, the scheme designer can anticipate that the demand can either surge, or be low, and thus act accordingly, an insight that an expected value deterministic plan would fail to provide. More importantly, the policy also informs the scheme designer that increasing supply primarily does the job of unlocking demand from Group 2. This is valuable as the designer can determine if the second group is important enough to be served and investigate reasons of Group 1's saturation.

4.4 Scenario 3: Bivariate Demand with a Bivariate Supply

The preceding scenarios have incrementally built the foundations of the FPD-OT framework. We now address the most comprehensive and increasingly common model in contemporary Sharing Economies (SEs): the allocation of resources from two distinct, potentially correlated, hybrid supply sources ($\mu(x_1, x_2)$) and competing demand groups ($\nu(y_1, y_2)$). The resulting FPD-OT plan, $\pi^O(x_1, x_2, y_1, y_2)$ is a 4D tensor.

4.4.1 Real World Imperatives for a 4D Approach

The scenario is common during demand surges, where SE designers combine platform-owned resources and peer-to-peer resources for diverse demand groups (e.g. price-sensitive vs inelastic). Examples (see Table 3) include Airbnb (Supply: whole houses and shared apartments; Demand: tourists and business visitors), clinic equipment supply to rural and urban areas, and distributed computing for academia and AI startups. The parking system research [8] (landlords and university staff competing for campus and residential parking) motivated this scenario.

4.4.2 Supply and Demand Marginals: $\mu(x_1, x_2)$, $\nu(y_1, y_2)$

Input Marginals to the FPD-OT programme, $\mu(x_1, x_2)$ and $\nu(x_1, x_2)$ are bimodal bivariate Poisson distributions. Here x_1 denotes the units of resource from P2P contributors whilst x_2 represents the platform-owned resources. These sources might be correlated as local regulations or events might affect both supplies.

$$\begin{aligned} \lambda_{s1a} = 5 \quad \lambda_{s2a} = 10 \quad \lambda_{s12a} = 1 \quad \lambda_{s1b} = 12 \quad \lambda_{s2b} = 10 \quad \lambda_{s12b} = 3 \quad p_s = 0.4 \\ \lambda_{d1a} = 8 \quad \lambda_{d2a} = 8 \quad \lambda_{d12a} = 2 \quad \lambda_{d1b} = 12 \quad \lambda_{d2b} = 10 \quad \lambda_{d12b} = 3 \quad p_d = 0.3 \end{aligned}$$

Figure 12 shows these marginals. As seen in $\mu(x_1, x_2)$, probability of an increased platform-owned supply is low, possibly due to operational difficulties.

4.4.3 Cost Metric: $C(x_1, x_2, y_1, y_2)$

A similar cost structure is adapted for bivariate supply. The new cost metric is a 4D tensor, Figure 13 shows slices of metric ($C(x_1, x_2, y_1^*, y_2^*)$ where y_1^* and y_2^* are demand realizations.

$$C(x_1, x_2, y_1, y_2) = c_o \max(0, (x_1 + x_2) - (y_1 + y_2)) + c_u \max(0, (y_1 + y_2) - (x_1 + x_2)) + c_t \max(0, (x_1 + x_2) - x_{\text{thresh}})$$

Table 4: Real-world scenarios of hybrid supply and competing demands

Name	$\mathbf{x}_1$: Primary Supply	$\mathbf{x}_2$: Secondary Supply	$\mathbf{y}_1$: Demand Segment 1	$\mathbf{y}_2$: Demand Segment 2
Ride-hailing	Platform-owned fleet	Peer-to-peer drivers	Daily commuters	Tourists
Airbnb Homestays	Entire homes	Shared apartments	Tourists	Business Travelers
Medical Equipment	Fixed hospital inventory	Mobile clinic units	Urban Patients	Rural Patients
Distributed Compute Platforms	Reserved Cluster Nodes	Spot Instances	Academic Research Jobs	Startups
Parking Spaces	University-owned spaces	Resident spaces	Staff	Locals
Bike-sharing	Docked bike stations	Dockless e-scooters	Daily commuter rides	Leisure rides
Electricity Grid	Base-load thermal plants	Flexible renewables/storage	Residential	Industrial

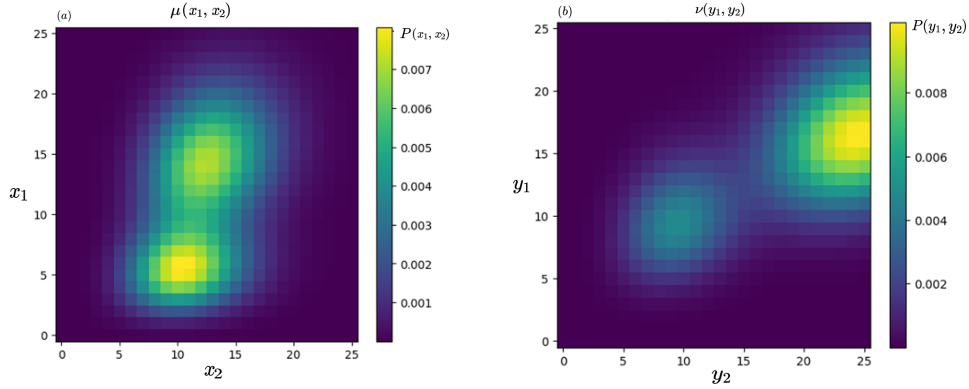


Figure 12: Input marginal probability distributions for Scenario 3: (a) Bimodal bivariate supply distribution $\mu(x_1, x_2)$, representing two distinct, correlated supply sources. (b) Bimodal bivariate demand distribution $\nu(y_1, y_2)$, representing two distinct, correlated demand groups.

4.4.4 Joint Optimal Plan: $\pi^O(x_1, x_2, y_1, y_2)$

The FPD-OT plan for the bivariate marginals, $\pi^O(x_1, x_2, y_1, y_2)$, is a 4D tensor showing optimal resource matching between both supply sources and demand groups. Given the challenge of visualizing and interpreting 4D tensors, conditional probabilities and induced marginals provide actionable insights for SE scheme designers. Figure 13 provides illustrative slices of these conditionals.

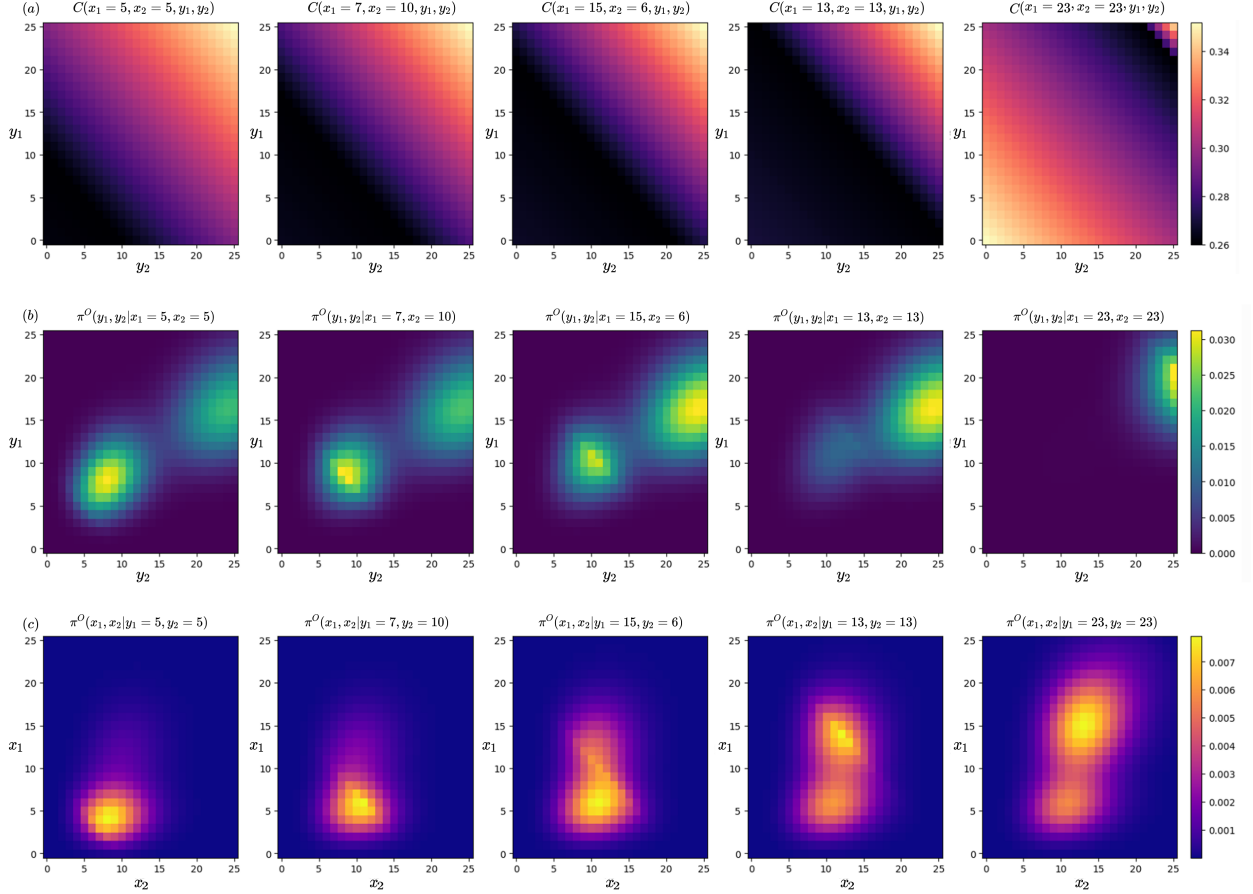


Figure 13: Slices from the 4D FPD-OT components for Scenario 3 (bivariate supply (x_1, x_2) , bivariate demand (y_1, y_2)): (a) Slices of the cost metric $C(x_1, x_2 | y_1^*, y_2^*)$, showing costs for fixed demand pair realizations (y_1^*, y_2^*) . (b) Conditional optimal demand distributions $\pi^O(y_1, y_2 | x_1^*, x_2^*)$, for fixed supply pair realizations (x_1^*, x_2^*) . (c) Conditional optimal supply distributions $\pi^O(x_1, x_2 | y_1^*, y_2^*)$, for fixed demand pair realizations (y_1^*, y_2^*)

Strategic Insights from Conditional Distributions: The complex relationship between the 2D marginals can be seen from supply and demand realizations.

- $\pi^O(y_1, y_2 | x_1, x_2)$: Helps optimally segment demand given specific supply. For weather-dependent car sharing with $x_1 = 7, x_2 = 10$ (Figure 4.4.4 (b) 2nd Heatmap), the conditional OT plan shows optimal target demand profiles of about 10 units for both groups. This is a combination of two things, the inherent demand probability distribution and the minimization of operational costs. The interesting aspect is to see the secondary mode at $y_1 = 16, y_2 = 24$. This suggests an alternative, though less probable, optimal allocation. If this scenario materializes, the total demand would be significantly higher ($16 + 24 = 40$ units), implying that the available supply of 17 units would lead to a substantial undersupply. The existence of this mode in the optimal plan suggests that even under-supplying to meet this demand profile might be preferable to other, even higher-cost outcomes, but also reflects the inherent uncertainty of the demand marginal $\nu(y_1, y_2)$. As expected, if the SE scheme is able to mobilize a high number of resources, the plan guides the designer to service a distinct mode with high values

of both demand groups.

- $\pi^O(x_1, x_2 | y_1, y_2)$: Generally, as total demand $(y_1 + y_2)$ increases, the plan recommends increased sourcing from both (x_1, x_2) . The specific shape and location of the high-probability regions in each plot highlight the nuanced sourcing strategy advised by the model. A key point to note is that the marginals selected, Figure 12 inherently would often lead to under-supply as $\mu(x_1, x_2)$ does not have a high probability of higher supplies. When the demand from both groups is forecasted to be 13 units, it is seen in the fourth conditional plan (Figure 13 (c)) that the program maintains a high probability at the line where $x_1 + x_2 = 26$, but due to the supply being stochastic, it also offers other contracts which might be optimal. The FPD-OT plan makes sure that it doesn't put mass in possible but expensive locations of the distribution.

Induced Supply-Demand Group Marginals from $\pi^O(x_1, x_2, y_1, y_2)$: While the FPD-OT Plan, $\pi^O(x_1, x_2, y_1, y_2)$, captures full relationships, and conditionals evaluate specific instances, underlying probabilistic relationships between specific supplies and demands are revealed by marginalizing (summing over two variables) along the plan. These help answer questions such as “*If the demand from group 1 is going to be (y_1) , how many resources of supply source 2 (x_2) should optimally be activated?*”

Figure 14 shows the four marginals derived from the two supplies and two demands. The brighter regions signify a higher probability of those pairings / contracts. Table 5 analyzes the marginals whilst offering an SE context.

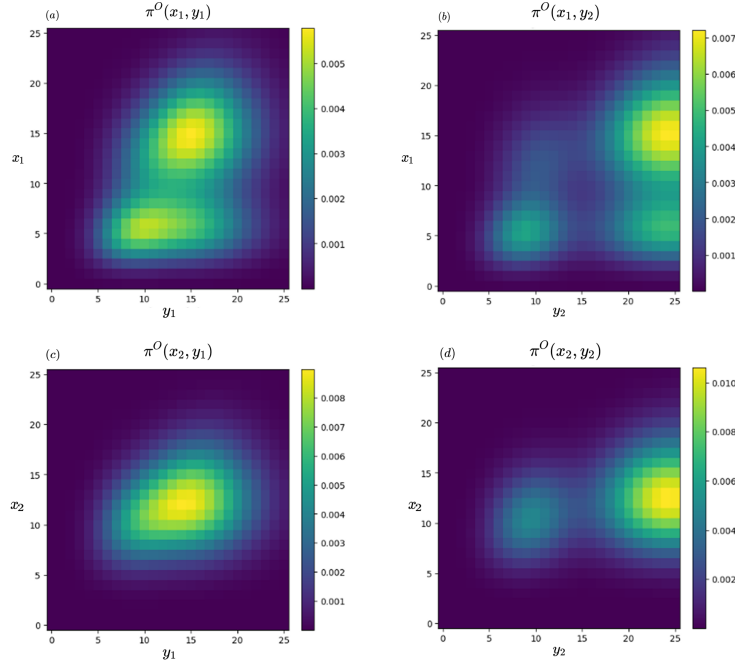


Figure 14: Induced 2D marginal probability distributions derived from the 4D optimal joint plan $\pi^O(x_1, x_2, y_1, y_2)$ in Scenario 3, revealing probabilistic relationships between specific supply-demand pairings: (a) $\pi^O(x_1, y_1)$; (b) $\pi^O(x_1, y_2)$; (c) $\pi^O(x_2, y_1)$; (d) $\pi^O(x_2, y_2)$

Table 5: Induced Supply–Demand Group Marginals from Optimal Plan $\pi^O(x_1, x_2, y_1, y_2)$

Marginal	Formula	Interpretation
Supply 1 x_1 and Demand 1 y_1	$\pi^O(x_1, y_1) = \sum_{x_2} \sum_{y_2} \pi^O(x_1, x_2, y_1, y_2)$	Mass concentrates around $x_1 = 6, y_1 = 10$ and $x_1 = 15, y_1 = 15$, signalling the two modes the scheme designer should look forward to matching. Neither of the clusters are significantly far off from the $x_1 = y_1$ diagonal, making sure that there is minimal under/over supply.
Supply 1 x_1 and Demand 2 y_2	$\pi^O(x_1, y_2) = \sum_{x_2} \sum_{y_1} \pi^O(x_1, x_2, y_1, y_2)$	A 4 modal heatmap for this combination reveals a complex state-dependent relationship rather than a simple linear correlation. The strongest mode is seen at a high demand and supply indicating the use of supply source 1 in order to meet surges in demand group 2. The presence of the other three weaker modes give the scheme designer a clear plan as to when to fulfil the demand group 2 with the first supply source.
Supply 2 x_2 and Demand 1 y_1	$\pi^O(x_2, y_1) = \sum_{x_1} \sum_{y_2} \pi^O(x_1, x_2, y_1, y_2)$	The distinct mode seen in this marginal at $x_2 = 13, y_1 = 14$ refers to a stable long term relationship between supply source 2 and demand group 1. The unimodality means that optimally, there is only one diffused state of having x_2 supplies from Supply 2 when there is a demand of y_1 from Demand 2.
Supply 2 x_2 and Demand 2 y_2	$\pi^O(x_2, y_2) = \sum_{x_1} \sum_{y_1} \pi^O(x_1, x_2, y_1, y_2)$	A bimodal marginal is seen in the last marginal as well with a strong mode on a high demand from Demand 2 and a moderate supply from Supply 2. Although, it can be noticed that an increase in y_2 should not lead to an increase in resources supplied from Supply 2. This is a key actionable insight which the scheme designer could take into consideration during demand surges.

5 QoS in a Fully Probabilistic Design Scenario

5.1 Quality of Service in SE

SE schemes thrive on two key notions: fairness and a reliable provision of services. Previous marginals analysis addressed group to group fairness. For reliability, fulfilling a minimum Quality

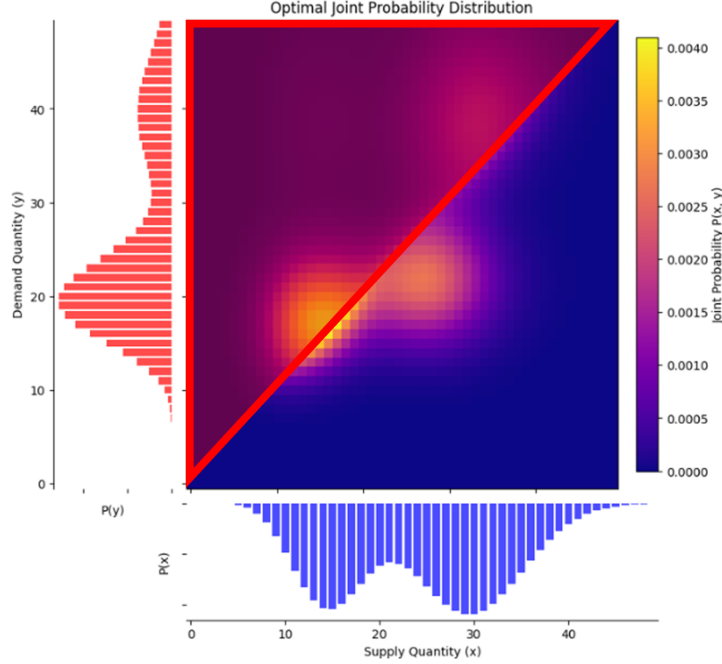


Figure 15: Conceptual visualization of the probability of undersupply for a univariate supply (x_1) and demand (y_1) scenario. The optimal joint plan $\pi^O(x_1, y_1)$ is shown as a heatmap, with the red triangular region representing all states where supply is less than demand ($x_1 < y_1$). The sum of probabilities within this region defines $P_{\pi^O}(\text{undersupply})$

of Service (QoS) ensures positive user perception. At most times, if a user requests a resource, they should be allocated one, minimizing undersupply. Section 4.2.3, mentioned that c_u (marginal cost of undersupply) would heavily penalise undersupply. Although optimal plans were generated, considerably high undersupply was still observed.

5.2 Probabilistic QoS in FPD-OT Scenarios

Whilst [8] and [7] use probability distributions to define QoS, a significant literature gap exists in defining QoS within FPD-OT joint plans. Until now, plans were primarily focused on cost-efficiency and satisfying marginals but the FPD-OT formulation allows to introduce new linear constraints. This dissertation proposes common QoS metric of undersupply to act as the third constraint of the problem, defined as:

$$\sum_{x_1 + x_2 < y_1 + y_2} \pi(x_1, x_2, y_1, y_2) = P_{\pi}(x_1 + x_2 < y_1 + y_2) \leq \alpha. \quad (10)$$

This constraint defines the probability of undersupply should be less than α . Figure 15 revisits the 1D scenario of section 4.2.5 and visualizes the total probability of undersupply for a 2D joint plan. This section explores two topics within QoS:

- **Feasibility Analysis:** Determining if adding a QoS constraint makes the problem infeasible
- **Sinkhorn-Knopp Modification:** A modified SK algorithm for feasible problems to satisfy marginals and the new QoS constraint computationally effectively.

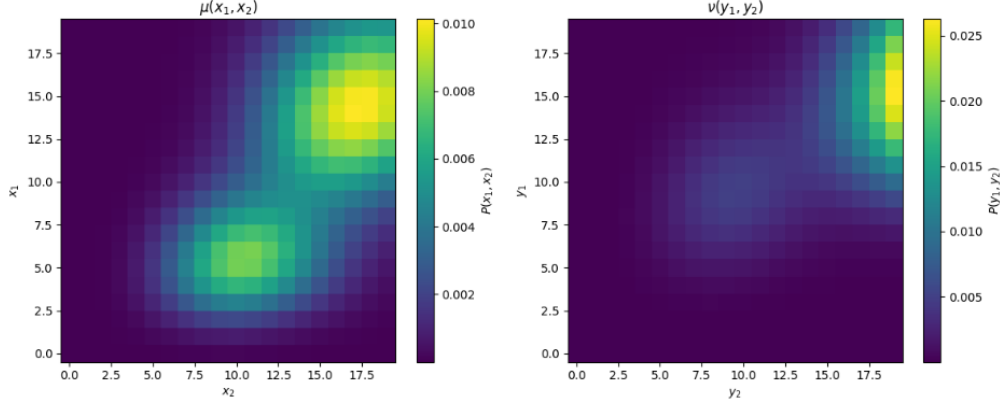


Figure 16: Bimodal bivariate marginal distributions used for the numerical example in the Quality of Service (QoS) analysis: (Left) Supply marginal $\mu(x_1, x_2)$; (Right) Demand marginal $\nu(y_1, y_2)$. These distributions are defined with $Q_{max} = 20$ per group for computational feasibility.

For realistic modeling, this section implements only the most complex scenarios: bivariate supply and demand.

5.3 Feasibility Analysis for QoS Guarantees

Designers requiring QoS guarantees with FPD-OT policies must first determine if the proposed QoS is feasible. The knowledge constraints (2, 3 and now the QoS constraint, 10), may yield an empty set if α is too restrictive.

5.3.1 Linear Program for α_{min}

To find the minimum probability of undersupply, α_{min} , below which no joint plan satisfying the QoS constraint exists, a linear program (LP) is used:

$$\alpha_{min} = \min_{\pi} \sum_{\text{s.t. } x_1 + x_2 < y_1 + y_2} \pi(x_1, x_2, y_1, y_2)$$

subject to:

$$\begin{aligned} \sum_{(y_1, y_2) \in \Omega_D} \pi(x_1, x_2, y_1, y_2) &= \mu(x_1, x_2) \quad \forall (x_1, x_2) \in \Omega_S \quad (\text{Supply Marginal Constraint}) \\ \sum_{(x_1, x_2) \in \Omega_S} \pi(x_1, x_2, y_1, y_2) &= \nu(y_1, y_2) \quad \forall (y_1, y_2) \in \Omega_D \quad (\text{Demand Marginal Constraint}) \end{aligned}$$

This feasibility LP does *not* involve the Ideal Distribution, $\pi^I(x_1, x_2, y_1, y_2)$ as the aim is to determine if a joint plan satisfying marginals and the QoS constraint can exist, not to minimize FPD-OT cost. The LP finds the plan with the minimum possible undersupply among all plans satisfying the marginals. If the scheme designer's desired QoS target is less than α_{min} , no such joint plan exists, requiring reconsideration of the QoS target.

For computational feasibility in the example, Q_{max} (maximum total supply/demand) is 20. Figure 16 shows the marginals used for the example, made with parameters:

$$\text{Supply } \mu(x_1, x_2) : \lambda_{s1a} = 5 \quad \lambda_{s1b} = 10 \quad \lambda_{s1ab} = 1 \quad \lambda_{s2a} = 12 \quad \lambda_{s2b} = 15 \quad \lambda_{s2ab} = 3 \quad p_s = 0.4$$

Demand $\nu(y_1, y_2) : \lambda_{d1a} = 8 \quad \lambda_{d1b} = 8 \quad \lambda_{d1ab} = 2 \quad \lambda_{d2a} = 12 \quad \lambda_{d2b} = 20 \quad \lambda_{d2ab} = 5 \quad p_d = 0.3$

The LP yields $\alpha_{min} = 0.2145$. No FPD-OT plan can guarantee QoS for α below this value.

5.4 Proposed SK Algorithm Modification

With α_{min} known, scheme designers choose a target QoS $\alpha \geq \alpha_{min}$, the FPD-OT framework now requires an algorithm to find the joint optimal plan, $\pi_{QoS}^O(x_1, x_2, y_1, y_2)$ satisfying knowledge constraints, whilst minimizing $KL(\pi || \pi^I)$. The dissertation proposes a modified SK algorithm with projection-normalization steps within each iteration to satisfy the QoS constraint.

5.4.1 4 Step Modified Sinkhorn-Knopp

Standard SK algorithm iteratively scales the ideal distribution $\pi^I(x_1, x_2, y_1, y_2)$ to match supply, $\mu(x_1, x_2)$ and demand marginals, $\nu(x_1, x_2)$. The modified algorithm at iteration k using plan $\pi^k(x_1, x_2, y_1, y_2)$ adds steps:

1. **Supply Marginal Scaling:** Update the plan to satisfy the supply marginal $\mu(x_1, x_2)$:

$$\pi^{(k*)}(x_1, x_2, y_1, y_2) = \pi^{(k)}(x_1, x_2, y_1, y_2) \cdot \frac{\mu(x_1, x_2)}{\sum_{y_1, y_2} \pi^{(k)}(x_1, x_2, y_1, y_2)}$$

for all (x_1, x_2, y_1, y_2) .

2. **Demand Marginal Scaling:** Update $\pi^{(k*)}$ to satisfy the demand marginal $\nu(y_1, y_2)$:

$$\pi^{(k**)}(x_1, x_2, y_1, y_2) = \pi^{(k*)}(x_1, x_2, y_1, y_2) \cdot \frac{\nu(y_1, y_2)}{\sum_{x_1, x_2} \pi^{(k*)}(x_1, x_2, y_1, y_2)}$$

for all (x_1, x_2, y_1, y_2) .

3. **QoS Constraint Projection:** Adjust $\pi^{(k**)}$ to satisfy the QoS constraint $P_\pi(x_1 + x_2 < y_1 + y_2) \leq \alpha$. First, calculate the current probability of undersupply:

$$P_{undersupply}^{(k**)} = \sum_{x_1 + x_2 < y_1 + y_2} \pi^{(k**)}(x_1, x_2, y_1, y_2)$$

If $P_{undersupply}^{(k**)} \leq \alpha$, then $\pi^{(k***)} = \pi^{(k**)}$ and repeat step 1. Otherwise if $P_{undersupply}^{(k**)} > \alpha$:

- For all (x_1, x_2, y_1, y_2) such that $x_1 + x_2 < y_1 + y_2$ (undersupply states):

$$\pi^{(k***)}(x_1, x_2, y_1, y_2) = \pi^{(k**)}(x_1, x_2, y_1, y_2) \cdot \frac{\alpha}{P_{undersupply}^{(k**)}}$$

- For all (x_1, x_2, y_1, y_2) such that $x_1 + x_2 \geq y_1 + y_2$ (non-undersupply states):

$$\pi^{(k***)}(x_1, x_2, y_1, y_2) = \pi^{(k**)}(x_1, x_2, y_1, y_2)$$

This operation ensures that in $\pi^{(k***)}$, the sum of probabilities for undersupply states is approximately α .

4. **Global Renormalization:** Since step 3 may alter the total sum of $\pi^{(k^{***})}$ (if $P_{\text{undersupply}}^{(k^{***})} > \alpha$), renormalize to ensure $\pi^{(k+1)}$ is a valid probability distribution:

$$\pi^{(k+1)}(x_1, x_2, y_1, y_2) = \frac{\pi^{(k^{***})}(x_1, x_2, y_1, y_2)}{\sum_{x_1, x_2, y_1, y_2} \pi^{(k^{***})}(x_1, x_2, y_1, y_2)}$$

This ensures $\sum_{x_1, x_2, y_1, y_2} \pi^{(k+1)}(x_1, x_2, y_1, y_2) = 1$.

These steps form one iteration. Repeat until convergence, when joint plan marginals are close to $\mu(x_1, x_2)$ and $\nu(y_1, y_2)$ and $P_\pi(x_1 + x_2 \leq y_1 + y_2) < \alpha$ is met.

5.5 Numerical Example

The modified algorithm is applied to Section 5.3.1 with parameters: $c_o = 1, c_u = 10, c_t = 5, x_{\text{thresh}} = 20, \epsilon = 0.1$. With $\alpha_{\text{min}} = 0.2145$, the scheme designer sets target QoS α to 0.4.

5.5.1 Non-QoS FPD-OT Plan

For the Non-QoS plan, we observe:

- Probability of Undersupply $P_{\pi^o}(x_1 + x_2 < y_1 + y_2) = 0.6592$
- Total Expected Cost $E_{\pi^o}[C] = 77.518$.
- Undersupply Cost = 47.310; Oversupply Cost = 1.013; Threshold Cost = 29.195.

This plan clearly violates the QoS constraint as $0.6592 > 0.4$, under-supplying more often.

5.5.2 QoS FPD-OT Plan

Applying the modified SK algorithm with target α yields a plan, $\pi_{QoS}^o(x_1 + x_2 < y_1 + y_2)$. The algorithm converged satisfying marginals and the QoS constraint within tolerance.

- Probability of Undersupply $P_{\pi_{QoS}^o}(x_1 + x_2 < y_1 + y_2) = 0.4000$
- Total Expected Cost $E_{\pi_{QoS}^o}[C] = 78.908$.
- Undersupply Cost = 47.573; Oversupply Cost = 1.140; Threshold Cost = 29.195.

5.5.3 Analysis of Change and Impact

The modified SK achieved the desired service level ($\alpha = 0.4$) reducing the total undersupply probability. However, this increased total expected cost from 77.5 to 78.9. The cost increase stems from a reduced feasible set due to the QoS constraint. Notably, expected undersupply cost increased even as its probability decreased. This suggests undersupplying less frequently but potentially with greater magnitude, impacting costs. Figure 17 compares conditional supply distributions from Non-QoS and QoS plans. The QoS plans shifts probability mass to reduce undersupply (i.e, towards $x_1 + x_2 \geq y_1 + y_2$) but must still satisfy marginals, leaving some probability in the undersupply region. Undersupply is more dispersed in the Non-QoS plan.

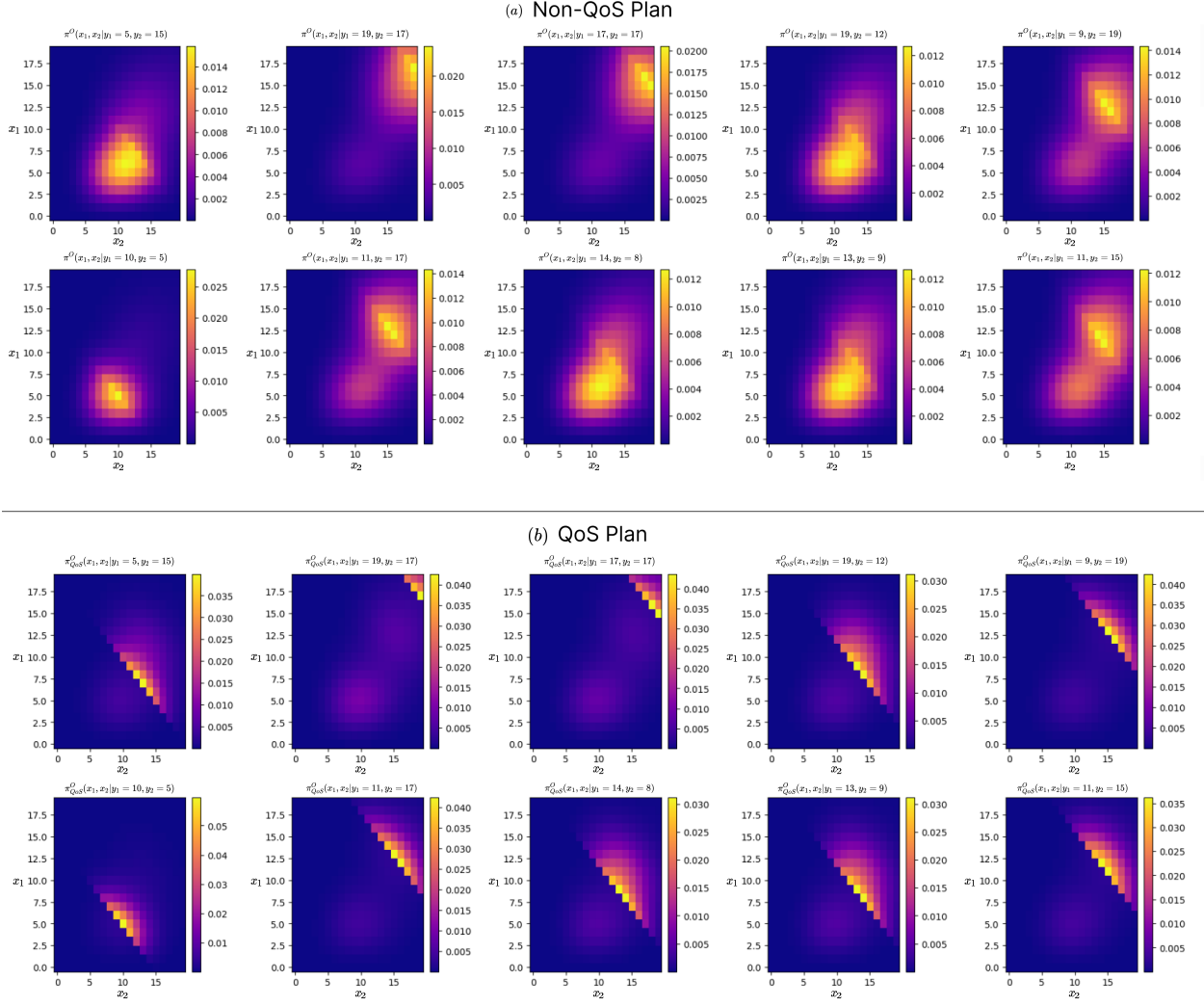


Figure 17: Slices of conditional optimal supply distributions $\pi^O(x_1, x_2 | y_1^*, y_2^*)$ for selected demand realizations (y_1^*, y_2^*) . The top row displays distributions from the standard FPD-OT plan (Non-QoS Plan), while the bottom row shows distributions from the FPD-OT plan incorporating an explicit undersupply probability constraint (QoS Plan), illustrating the impact of the QoS constraint on allocation strategy.

5.6 Implication for Scheme Designers

The novel integration of QoS constraints in FPD-OT framework can be extended in multiple ways. While undersupply probability is a common QoS metric, designers can target others, like oversupply or specific supply-demand source pairings. Designers can move beyond indirect cost minimization via a high c_u to set explicit probabilistic service targets. The feasibility test reveals potential for reducing undersupply to meet policies/regulations, even if a current plan is cost-efficient. Probabilistic QoS directly targets long-term user satisfaction and platform trust. This method can also help introduce service tiers, each minimizing cost subject to specific QoS guarantees.

6 Evaluation with State of the Art Allocation Strategies

6.1 Summary of the Proposed FPD-OT Framework

To assess the FPD-OT approach’s efficacy for SE resource redistribution, a comparative evaluation against state-of-the-art strategies (from Section 2.2) is designed. The evaluation focuses on the most complex scenario: bivariate supply and demand (Section 4.4). The proposed FPD-OT framework provides designers with probabilistic allocation policies, offering deeper risk assessment than single-value recommendations. For quantifiable performance comparison, these probabilistic plans are distilled into specific FPD-OT derived allocation strategies to benchmark against existing methods.

6.2 Benchmarking Methodology

A Monte Carlo simulation mimics many “operational days”, with demand pairs (y_1, y_2) drawn from $\nu(y_1, y_2)$. For evaluating allocation policies (which determine supply given demand/historical patterns), benchmark policies set supply (x_1, x_2) using their internal logic. FPD-OT strategies use the conditional distribution $\pi^O(x_1, x_2|y_1, y_2)$ to decide a supply pairing.

The performance of the strategies is assessed by average daily operational cost (defined in Section 4.4.3) over the total simulation period. Monte Carlo input parameters are mentioned in Table 6.

Table 6: Simulation Input Parameters	
Parameter	Value
Number of Simulation Days	100,000
Q_{max}	26 (0 to 25)
$\mu(x_1, x_2)$ Parameters	
Mode 1 ($\lambda_{s1a}, \lambda_{s1b}, \lambda_{s1ab}$)	(11, 6, 1)
Mode 2 ($\lambda_{s2a}, \lambda_{s2b}, \lambda_{s2ab}$)	(20, 12, 1)
p_s	0.5
$\nu(y_1, y_2)$ Parameters	
Mode 1 ($\lambda_{d1a}, \lambda_{d1b}, \lambda_{d1ab}$)	(5, 2, 1)
Mode 2 ($\lambda_{d2a}, \lambda_{d2b}, \lambda_{d2ab}$)	(14, 20, 2)
p_d	0.4
Cost Parameters	
c_o	1
c_u	10
x_{thres}	20
c_t	5
FPD-OT Parameters	
ϵ	0.1

6.3 Benchmark Allocation Strategies

6.3.1 Static Mean Supply (SMS)

Sets fixed supply level ($E[x_1], E[x_2]$) derived from $\mu(x_1, x_2)$, ignoring demand distribution entirely.

6.3.2 Total Mean Match (TMM)

Matches average total demand, $E(y_1 + y_2)$, splitting into x_1, x_2 based on the expected supply ratio from $\mu(x_1, x_2)$.

6.3.3 Newsvendor

Calculates an optimal total supply level and then splits it based on the expected supply pairing from $\mu(x_1, x_2)$. The optimal total supply is found by [10]:

1. Calculating the critical fractile, $CF = c_u / (c_o + c_u)$ using the cost coefficients.
2. Finding the smallest total supply quantity $(y_1 + y_2)$ whose cumulative probability in the total demand distribution (derived from $\nu(y_1, y_2)$), meets CF.

6.3.4 Sample Average Approximation (SAA)

This probabilistic method considers a wide range of possible supply pairs, for which it generates $N_{samples}$ demand realizations from $\nu(y_1, y_2)$. It calculates the cost of the candidate supply pairs against the demands and chooses the single (x_1, x_2) that has the lowest average cost [11].

6.4 FPD-OT Derived Strategies

Four FPD-OT derived strategies, based on the conditional plan, $\pi^O(x_1, x_2 | y_1, y_2)$, are evaluated: using its Mean, Mode, a random Sample, or a Top-K selection (details in Table 7).

Table 7: FPD-OT Derived Allocation Strategies

Strategy	Description / Decision Logic	Mathematical Formulation / Basis
Mean	Uses the expected value of each supply component from the conditional distribution $\pi^O(x_1, x_2 y_1, y_2)$ for a given demand realization (y_1, y_2) .	$x_1^* = \sum_{x_1, x_2} x_1 \pi^O(x_1, x_2 y_1, y_2)$ $x_2^* = \sum_{x_1, x_2} x_2 \pi^O(x_1, x_2 y_1, y_2)$
Mode	Selects the supply pair (x_1^*, x_2^*) that has the highest probability in the conditional distribution $\pi^O(x_1, x_2 y_1, y_2)$ for the observed demand pair.	$(x_1^*, x_2^*) = \underset{(x_1, x_2)}{\operatorname{argmax}} \pi^O(x_1, x_2 y_1, y_2)$
Sampling	Randomly samples a supply pair (x_1^*, x_2^*) from the conditional distribution $\pi^O(x_1, x_2 y_1, y_2)$ for the current demand realization.	$(x_1^*, x_2^*) \sim \pi^O(x_1, x_2 y_1, y_2)$
Top-K	Adapted by how LLMs pick their next token [33], it identifies the K supply pairs (x_1, x_2) with the highest probabilities in $\pi^O(x_1, x_2 y_1, y_2)$. From this set, selects the pair that minimizes the predefined cost function $C(x_1, x_2, y_1, y_2)$.	1. Select Top-K pairs from $\pi^O(x_1, x_2 y_1, y_2)$. 2. $(x_1^*, x_2^*) = \underset{(x_1, x_2) \in \text{Top-K}}{\operatorname{argmin}} C(x_1, x_2, y_1, y_2)$.

6.5 Results and Analysis

Key results (Table 8) show FPD-OT derived strategies achieve superior performance (lower average daily cost) compared to benchmarks.

- **FPD-OT Top-K** (Avg Cost: 53.83, Std Dev: 46.98): The top performer, this hybrid approach effectively navigates the 4D cost landscape with relatively consistent performance (lowest Std Dev).
- **FPD-OT Mode** (Avg Cost: 55.44, Std Dev: 48.21): Second-best performer, its strength suggests that the conditional plan’s mode often aligns with low-cost outcomes.

Table 8: Summary of Simulation Results: Average Daily Cost and Standard Deviation

Policy	Avg Cost	Std Dev
FPD-OT (Cond. Mean)	70.0320	58.4426
FPD-OT (Cond. Mode)	55.4361	48.2137
FPD-OT (Sampling)	80.5206	62.6250
FPD-OT Top-K	53.8313	46.9773
Total Mean Match (TMM)	96.6567	60.3877
Newsvendor (Total)	123.6191	14.1467
Sample Average Approx. (SAA)	95.7125	48.2327
Static Mean Supply (SMS)	96.2371	56.2221

- **FPD-OT Mean** (Avg Cost: 70.03, Std Dev: 58.44): While taking the expected value seems intuitive, it might fall short in multimodal or skewed scenarios leading to supply decisions that are not necessarily optimal point.
- **FPD-OT Sampling** (Avg Cost: 80.52, Std Dev: 62.63): Highest FPD-OT average cost and standard deviation. Single random samples can be suboptimal, especially if drawn from less cost-effective tails of the conditional distribution, accumulating higher costs over time. Although, the long run induced supply distribution matched $\mu(x_1, x_2)$, signifying long term fairness and realism.

6.6 Commentary on Other Benchmark Strategies

Other benchmarks generally underperformed against FPD-OT approaches.

- **Static Mean Supply (SMS)** (Avg Cost: 96.24, Std Dev: 56.22): Its fixed supply (16, 10) often mismatched with daily demand fluctuations, causing high costs from under/oversupply due to ignoring demand information.
- **Total Mean Match (TMM)** (Avg Cost: 96.66, Std Dev: 60.39): Performed similarly to SMS. Though slightly more demand aware, its reliance on expected values, rather than full distributional shape, proved insufficient.
- **Sample Average Approximation (SAA)** (Avg Cost: 95.71, Std Dev: 48.23): Despite pre-calculating an optimal fixed supply (3, 25) from $N_{samples} = 1000$ demand samples, SAA yielded high average costs. This chosen supply pair (3, 25) appears to be a specific point for the sampled demands but does not generalize well to the full range of demand scenarios. The decision to supply low x_1 (3) and very high x_2 (25) is notably different from other policies, producing a less balanced approach. This raises fairness questions for deterministic policies as group 1 would be consistently under-supplied.
- **Newsvendor** (Avg Cost: 123.62, Std Dev: 14.15): Performed the worst by average cost, likely due to over-reliance on the cost parameters. Its low standard deviation suggests consistently suboptimal high-cost decisions.

FPD-OT strategies demonstrated a clear advantage by conditioning their probabilistic supply plan on observed daily demand in this complex, stochastic environment. The FPD-OT Top-K strategy, in particular, shows that a sophisticated interpretation of the FPD-OT joint plan can yield the most cost-effective resource allocation. It is to be reiterated that the dissertation proposes the use of the whole distribution rather than single-value contracts.

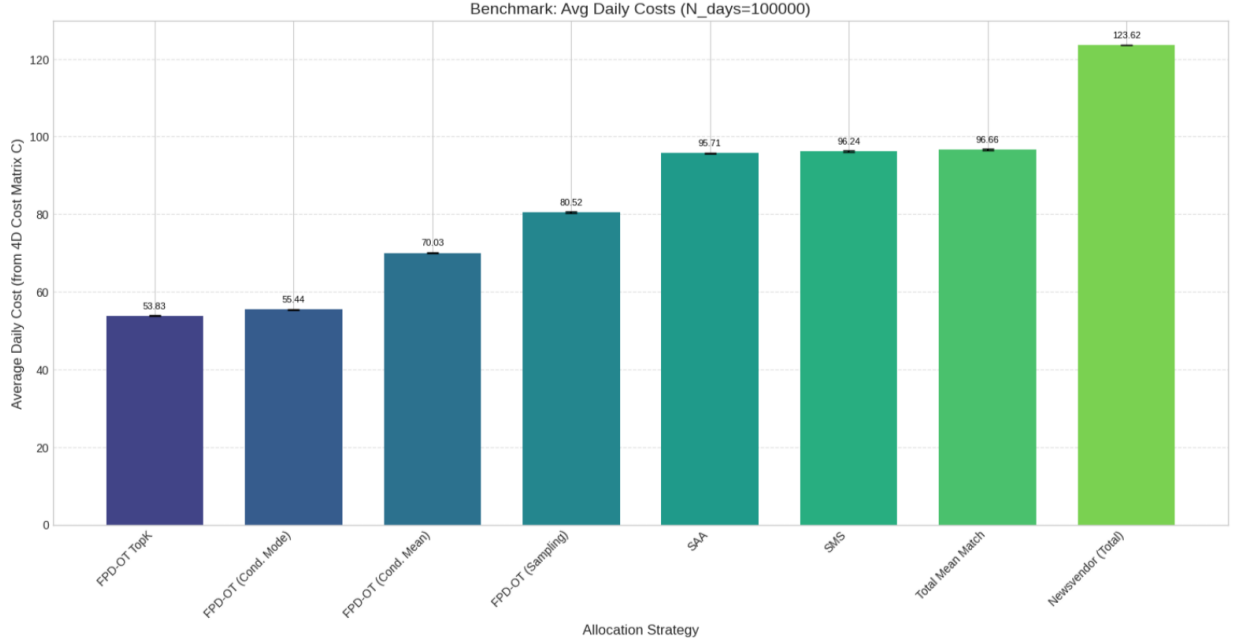


Figure 18: Bar chart summarizing the average daily operational costs of various allocation strategies evaluated over 100,000 simulated days for Scenario 3. Strategies are arranged in ascending order of mean cost, comparing FPD-OT derived policies against benchmark methods.

7 Discussion and Conclusion

This dissertation has addressed the complex resource redistribution challenges inherent in contemporary multivariate SE schemes by proposing a novel framework using FPD-OT and evaluating it against benchmark strategies. This section will talk about the key contributions of the dissertation, related explorations of FPD, its tradeoffs and limitations, and outlining directions for future work.

7.1 Contributions of the Dissertation

1. **Proposal of FPD-OT Framework for SE Scheme Designers:** Traditional methods in SE design fall short to take the stochastic nature of supply and demand and abstracts away distributional information. The proposed approach provides designers with a powerful tool that represents an optimal policy which can be analyzed in multiple ways (e.g. conditionals, marginals, etc) in order to gain both short and long term actionable insights. The joint plan, $\pi^O(x_1, x_2, y_1, y_2)$, makes sure that the scheme runs on minimal costs whilst satisfying the ex-ante distributional information for both supply and demand.
2. **Give SE Context to FPD-OT Objects:** The dissertation bridges the theoretical objects of FPD-OT with the practical realities of SE schemes operations. By translating abstract mathematical notations into tangible levers and interpretable components for scheme designers, the framework becomes more accessible and actionable. Table 9 details these contextualizations.
3. **Integration of QoS in FPD-OT Programme:** Mapping a conventional mindset about QoS into FPD-OT scenarios was a key achievement. This was achieved through two key developments:

Table 9: Sharing Economy Contextualization of Mathematical Tuning Knobs of FPD-OT

Mathematical Object	SE Context
Cost Parameters $(c_o, c_u, c_t, x_{thresh})$	The metric and its parameters are open to be expressed by the designer based on their scenario. The economic penalty allows the FPD-OT programme to determine the ideal distribution which has zero loss. The general cost metric introduced in this research takes the three most important cost components of the real world. A high c_u, for instance, commands the system to prioritize service availability, even if it means risking some oversupply.
Regularization Parameter (ϵ)	Acting as “temperature”, this tuning knob decides if the plan should focus more on the financial cost or the smoothness, allowing a broader range of good contracts. A smoother plan might not be the most financially optimal, but stands against fluctuations in the market.
Base Distribution (ϕ)	The belief is often used to inject prior knowledge about the system, if certain pairings are more desirable. The dissertation assumed it as a uniform distribution for simplicity but it can be a powerful tool to drive the FPD-OT joint plan to a certain shape.
Supply and Demand Marginals $(\mu(x_1, x_2), \nu(y_1, y_2))$	The historical (or forecasted) supply and demand distributions are the “Knowledge Constraints” or realities that the plan must respect. These marginals can either be modelled based on certain known data points (like in this dissertation) or can simply be a distribution of historical supply and demand to get realistic operational context. Modelling the distributions also allows for stress testing to understand how can costs be further minimized.
Quality of Service (QoS) and Target (α)	A number of constraints can be added to the FPD program to define QoS, allowing the designer to determine the non-negotiables. In context of undersupply, α acts as the minimum allowed undersupply the designer is willing to handle. This transforms a qualitative service objective into a quantifiable, enforceable constraint within the FPD-OT optimization, compelling the system to adhere to a defined level of service reliability.
Joint Plan $(\pi^O(x_1, x_2, y_1, y_2))$ and Strategies	The key output of the FPD-OT programme, its structure (e.g., modes, spread, correlations revealed through conditional probabilities and induced marginals) provides important insights into optimal resource flows, risk concentrations, and inter-group dynamics. The choice of allocation strategies such as utilizing the conditional mean, conditional mode, random sampling from the conditional distribution, or the more nuanced Conditional Top-K approach determines how the probabilistic guidance is translated into specific supply actions, thereby influencing the realized operational performance.

- **QoS Feasibility Analysis:** The LP formulation to determine the minimum achievable probability of undersupply (α_{min}) given the system’s inherent supply and demand marginals. This provides an upfront assessment for designers regarding the viability of desired service levels.
- **Modified Sinkhorn-Knopp Algorithm:** The development and empirical validation of a modified Sinkhorn-Knopp algorithm. This iterative procedure is designed to converge to an optimal plan $\pi_{QoS}^O(x_1, x_2, y_1, y_2)$ that satisfies the marginal constraints, QoS

constraint, and minimizes KL divergence from the ideal plan. This represents a methodological advancement, allowing for direct enforcement of guarantees within the FPD-OT optimization, rather than relying on cost penalties to indirectly influence service outcomes.

7.2 Tradeoffs and Limitations of FPD-OT for SE

Despite its advantages, the FPD-OT framework, in its current form, presents certain limitations that might need consideration. A primary concern is the computational scalability with dimensionality as the number of supply and demand groups increases, the size of the joint plan tensor grows with it. While the SK algorithm is efficient for a given problem size, its application to very high-dimensional SE scenarios may require increased computation or more modifications. While analytical tools are provided, the interpretability of high-dimensional plans (e.g., 4D or higher) can be unintuitive and demanding, underscoring the need for better visualization and analytical techniques to leverage the information within complex probabilistic plans. Further, the current implementation relies on static supply and demand marginals ($\mu(x_1, x_2)$ and $\nu(y_1, y_2)$), derived from historical data. These distributions can be dynamic. A plan optimized for one set of marginals may lose optimality if these underlying distributions shift significantly. Further, the plan is also primarily trying to minimize $C(x_1, x_2, y_1, y_2)$ which might not always be nuanced enough to capture all real life costs and objectives.

7.3 Future Research Directions

The contributions and limitations discussed highlight several promising next steps to extend the dissertation’s proposal. The motivation of the paper was the parking SE scheme presented in Section 4, an example of a nuanced multidimensional problem. This research has set the foundation to convert that scheme into a probabilistic design too. Exploration into more granular, group-specific, and dynamic cost structures could lead to more nuanced and fair allocation strategies. For QoS, it is vital to find analytical solutions to the feasibility test instead of the LP to deal with large datasets. Some early analysis into Fréchet-bounds was done whilst preparing the code and can be extended. Further, providing a formal convergence proof for the modified Sinkhorn-Knopp algorithm is of utmost importance to solidify its robustness. Once achieved, the scope of service guarantees can be expanded to a diverse set and can also be used in policy design.

7.4 Conclusion

The project has successfully tackled the research challenge of incorporating probabilistic decision making in SE schemes and has proven its advantages with a thorough evaluation section. With the QoS incorporation, the dissertation has gone beyond the state of the art to propose more cost-effective and user satisfactory policies.

8 Project Management and Reflection

8.1 Timelines and Project Scoping

The project adhered to the timelines it had initially set out with successfully updating at both Early and Late Stage gateways. The initial aim of the project was less focused on complex SE schemes, and more on the notion of fairness in SE. Irrespectively, little work was made redundant as learnings from the past models deeply helped in evaluating new solutions. At a stage, the dissertation proposed the design of supply PMFs rather than finding joint plans, but was soon discontinued to stick to the traditional OT aims, matching supply *with* demand. Once the clear objective of the 2D 2D scenario was set, the problem was broken down into three stages (ref. Table 10), with incremental complexity to scope the dissertation. Each stage helped make the next relatively easier to tackle.

Table 10: Project Scenarios and Focus

Scenario No.	Description	Key Focus/Development
1	Univariate Supply and Univariate Demand	Develop foundational FPD-OT tools used throughout the dissertation; explore multimodal transport plans.
2	Univariate Supply and Bivariate Demand	Build tools and code to obtain and visualise multivariate plans.
3	Bivariate Supply and Bivariate Demand	Collate insights; highlight the role of conditionals and marginals in the final joint plan.

8.2 Resource Management

Regular meetings with supervisor allowed for a constant check if the project was headed in the correct direction. Project resources, including Python code developed on Google Colab and a comprehensive Overleaf document serving as a regularly updated logbook of findings and results since October 2024, were managed in a structured and auditable manner. The PDF of the logbook and a snapshot of an entry can be found in Appendix C.

8.3 Reflection

This project significantly advanced my skills in probabilistic modelling (FPD-OT), algorithm design, and academic research, aligning with my interests in optimization and complex systems. Picking up on modules like Optimization and ML for Design Engineers, allowed me to deal comfortably with unknown mathematical concepts but also gave the confidence to work on a technically challenging project. Addressing 4D tensors and novel QoS integration pushed my capabilities, identifying areas for future growth in large-scale computation.

Whilst working on the project, I took upon multiple coding challenges (e.g. Bivariate Gaussian Modelling) to build my foundations on tackling the harder problems. The FPD-OT framework is an original response to SE resource allocation, refined through an iterative process involving scope adjustments and a staged scenario analysis, culminating in a robust solution. Appropriate methods and tools, including Python, FPD/OT principles, and custom algorithms (necessitated

by standard library limitations), were skillfully applied. This ensured the project's technical depth and successful achievement of its objectives, tackling complex challenges with a well-reasoned and creative approach.

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9 Use of Generative AI

This research involved some use of Generative AI tools. It is to be confirmed that all work done, including the report is original and is in no part AI generated. Following is the list of the AI tools and their use:

1. Perplexity: Used to find sources relating to certain research topics. During the literature review phase, conversations were used primarily to lead to academic papers for a searched domain.
2. ChatGPT: Primarily used to grasp foundational probability theory concepts and to perform sanity checks on certain code files.
3. Google Gemini: Used as a coding copilot in order to fix syntax, find inefficiencies, commenting and improving code readability.

A Construction of Bivariate Bimodal Poisson Distributions

This appendix outlines the construction of the bivariate bimodal Poisson distributions used to model supply and demand marginals, corresponding to $\mu(x_1, x_2)$ and $\nu(y_1, y_2)$ in Scenario 3.

A.1 Univariate Poisson Distribution

The standard univariate Poisson Probability Mass Function (PMF) is:

$$P(K = k|\lambda) = \frac{e^{-\lambda} \lambda^k}{k!}$$

where:

- k : number of occurrences ($k = 0, 1, 2, \dots$).
- λ : mean parameter.

A.2 Correlated Bivariate Poisson Distribution (Single Mode)

A single mode of a correlated bivariate Poisson distribution is constructed using three independent Poisson variables $Z_A \sim \text{Poisson}(\lambda_A)$, $Z_B \sim \text{Poisson}(\lambda_B)$, and $Z_{AB} \sim \text{Poisson}(\lambda_{AB})$. The correlated variables are $K_1 = Z_A + Z_{AB}$ and $K_2 = Z_B + Z_{AB}$. The PMF is:

$$P(K_1 = i, K_2 = j|\lambda_A, \lambda_B, \lambda_{AB}) = e^{-(\lambda_A + \lambda_B + \lambda_{AB})} \sum_{l=0}^{\min(i,j)} \frac{\lambda_A^{i-l}}{(i-l)!} \frac{\lambda_B^{j-l}}{(j-l)!} \frac{\lambda_{AB}^l}{l!}$$

The parameters $\lambda_A, \lambda_B, \lambda_{AB}$ for each mode and supply/demand type are denoted as λ_{kMc} in the dissertation's notation table, where:

- $k \in \{s, d\}$: Identifies Supply or Demand.
- $M \in \{1, 2\}$: Identifies the specific supply (X_M) or demand (Y_M) group. For mode $c \in \{a, b\}$, the parameters map as:
 - $\lambda_A \rightarrow \lambda_{kMa}$ (mean for the first variable of the pair in mode a)
 - $\lambda_B \rightarrow \lambda_{kMb}$ (mean for the second variable of the pair in mode a)
 - $\lambda_{AB} \rightarrow \lambda_{kMab}$ (covariance term for the pair in mode a)

For example, for supply sources X_1, X_2 and its first mode (denoted 'a' in the general notation $\lambda_{s1a}, \lambda_{s1b}, \lambda_{s1ab}$), these correspond to the code's `lambda1_s1`, `lambda2_s1`, `lambda12_s1`. Similar mappings apply for other modes and for demand. Two such modal PMFs, $P_k^{\text{mode } a}(i, j)$ and $P_k^{\text{mode } b}(i, j)$, are computed.

A.3 Bimodal Bivariate Poisson Distribution

The final bimodal PMF for supply (P_s) or demand (P_d) is a weighted sum of its two modal PMFs:

$$P_k(i, j) = p_k \cdot P_k^{\text{mode } a}(i, j) + (1 - p_k) \cdot P_k^{\text{mode } b}(i, j)$$

where p_k is the mixture weight for distribution $k \in \{s, d\}$.

- For supply, p_s is `w_s` in the code (e.g., 0.4 for Scenario 3).
- For demand, p_d is `w_d` in the code (e.g., 0.3 for Scenario 3).

The resulting $P_s(i, j)$ and $P_d(i, j)$ are normalized and serve as the input marginals $\mu(x_1, x_2)$ and $\nu(y_1, y_2)$ respectively. Parameter values for Scenario 3 are detailed in Section 4.4.2.

B Python Google Colab Codebase

The code for all 3 scenarios, the feasibility test, modified SK algorithm, along with the evaluation Monte Carlo can be found on the following Google Colabdrive link:

https://colab.research.google.com/drive/1N0p4mtvMooeMc_B2ZtbhWrpVsd1k304S?usp=sharing

C Master's Project Logbook

Each meeting with supervisor was recorded on the logbook and any progress made was uploaded as well. Every entry is time-stamped. <https://drive.google.com/file/d/1VMkplOZiC-D-bhh6sVYCz23mCQRQkkmL/view?usp=sharing>

14/01/2025

1. Discussed work done over the break:
 - (a) Fixing GMM code and reviewing the geometric combination. Talked about precision and how it is linearly related to the weights. Can we rewrite the slightly complicated way of writing the mean with betas instead. [Review the important idea of a convex combination, which applies here](#)
 - (b) Spoke about how it is important to embed the results in a narrative
 - (c) Discussed the other combinations (barycentric and wasserstein [What's the difference / meanings here?](#)). Discussed the 2 step optimization process in OT. [I'll send you some slides on this](#)
 - (d) Went over the KLD vs. fixed parameters to understand interesting relationships. [Not sure what this entry means. There was a discussion about how some of your graphs \(ordinates\) need to be relabelled to emphasize that they are min quantities](#)
2. Make a bib file to keep note of all references and citations [very important](#). I need to be able to navigate [the working draft to answer queries such as "what literature has Aditya reviewed on the deployment of OT methods in SE?"](#) etc
3. Key task now is to start working on problem 2. Start writing down an analysis and description to solve it. Spoke about considering the demands as Poisson distributions. [Most pressing task now](#)
4. Discussed the list of possible applications and start finding interesting data and realistic applicability for the scope of the project. [Your simulation-based investigations \(MC studies\) are scientifically important, but there's no doubt that linking to a real problem, with real data, and showing a benefit \(additivity\) would really put wind in the sails of the project](#)

Figure 19: Snapshot of a typical logbook entry made on Overleaf for the research project.